

## 5 Dab in the North Sea, Skagerrak, and Kattegat

### dab.27.3a4 – *Limanda limanda* in Subarea 4 and Division 3.a

#### 5.1 General

Dab (*Limanda limanda*) was assessed for the first time by the Working Group on the Assessment of Demersal Stocks in the North Sea and Skagerrak (WGNSSK) in 2014 (ICES, 2014). Until 2013, dab was assessed by the Working Group on Assessment of New MoU Species (ICES, 2013). This group was dissolved in 2014. Because only official landings and survey data were available at that time, dab was defined as a category 3 species according to the ICES guidelines for data limited stocks (ICES, 2012). Since 2015, dab was included in the official data call for the WGNSSK and discard estimates could be included into the dab assessment since then. In 2016, a benchmark assessment of dab was conducted by ICES. For this benchmark assessment, catch data from 2002 were requested and uploaded into the InterCatch data portal by all relevant countries (ICES, 2016). The benchmark agreed on the use of a survey-based assessment model (SURBAR; Needle, 2015) to inform stock status of North Sea dab (ICES, 2016). This model provides relative estimates of the spawning stock, recruitment, and estimates of total mortality. During the WGNSSK 2017,  $MSY_{proxy}$  reference points were determined applying the Surplus Production Model in Continuous Time (ICES, 2017b; SPiCT, Pedersen and Berg, 2017), and catch advice for dab was provided for the years 2017–2019 applying the 2:3 rule based on the SURBAR spawning stock biomass index. In 2017, the combined TAC for dab and flounder was removed (EU COM, 2017/595). North Sea dab has become a non-target species with no TAC since then, and ICES has not been requested to provide advice on fishing opportunities for this stock for the years 2020–2022. In 2022, catch advice for the dab stock was requested again for the years 2023–2025. Catch data, indices and the SURBAR assessment were updated, and also the SPiCT assessment was updated during the WGNSSK 2022. However, the existing SPiCT assessment was not accepted as the basis of the advice, due to unacceptable strong patterns in the retrospective analysis. Therefore, the Length Based Indicator was used to evaluate stock status relative to the  $F_{MSY proxy}$ , and the empirical *chr* rule (ICES, 2021a) was applied as the basis of advice. During the WGNSSK 2025 catch data, indices and the SURBAR assessment were updated, and a new catch advice was prepared for 2026, 2027, and 2028 based on the *chr* rule. For 2024, catches were estimated at 23 174 tonnes (compared to 28 005 tonnes in 2023). The relative SSB value decreased slightly. Recruitment, as estimated from the SURBAR assessment, showed a decreasing trend from 2015 to 2020, fluctuated since on medium levels, but decreased in 2024. The stock status was checked by the LBI method, which revealed that the exploitation status was below the  $F_{MSY proxy}$ . The advice based on the *chr* rule is as follows: „ICES advises that when the MSY approach is applied, catches in each of the years 2026, 2027 and 2028 should be no more than 19 767 tonnes. If discard rates do not change from the average of the last three years (2022–2024), this implies landings of no more than 1 313 tonnes“

##### 5.1.1 Biology and ecosystem aspects

Dab is a widespread demersal species on the Northeast Atlantic shelf and distributed from the Bay of Biscay to Iceland and Norway, including the Barents Sea and the Baltic Sea. In the North Sea it is one of the most abundant species distributed over the whole area in depths down to 100 m, but it was also found occasionally down to depths of 150 m. The main concentration of dab (especially the younger age groups 1–2) can be found in the southeastern North Sea. Older

age groups occur more in the central and more Northern parts of the North Sea (Figure 5.14). Generally, dab abundance decreases towards the northern parts of the North Sea. Dab feeds on a variety of small invertebrates, mainly polychaete worms, shellfish and crustaceans. Early sexual maturation was reported for dab, maturing at ages of 2 to 3 years corresponding to approximately 11 cm to 14 cm total length. Peak spawning in the southeastern North Sea occurs from February to April.

### 5.1.2 Stock ID and possible assessment areas

The several spawning grounds and the wide distribution of dab indicate the presence of more than one stock. Meristic data (Lozán, 1988) corroborate the hypothesis of several stocks for dab, distinguishing significantly between populations from western British waters, the North Sea and the Baltic Sea.

### 5.1.3 Management regulations

Dab is mainly a bycatch species in fisheries for plaice and sole. The discard rates for dab can be extremely high (~90%). No minimum landing size is defined for dab on an EU level. According to EU-Regulations a precautionary TAC was given in EU waters of Division 2.a and Subarea 4 together with flounder (*Plathichthys flesus*). This combined TAC was never fully utilized. In 2017, the European Commission requested ICES to evaluate the possible effects on the stocks of dab and flounder having no TAC. ICES advised that given the current fishing patterns of the main fleets catching dab and flounder, which are the same fleets targeting plaice and sole, the risk of having no TAC for dab and flounder is considered to be low (ICES, 2017a). Therefore, the European Commission removed the combined TAC for these two stocks in 2017 (EU COM, 2017/595).

## 5.2 Fisheries data

### 5.2.1 Historical landings

Dab is a bycatch species mainly in the fisheries for plaice and sole but also in fisheries targeting demersal round fish. According to official catch statistics, annual landings of dab in ICES Subarea 4 and Division 3.a have been increasing above 10 000 tonnes since 1979 (Figure 5.1–5.3, Table 5.6 – 5.8). The apparent decrease in official landings in the 1980s and 1990s are due to unreported landings by the Netherlands. However, since 1999, total landings for both areas (Subarea 4 and Division 3.a) steadily decreased. This trend continued until 2017 with total official landings of 3918 tonnes. In 2022, the official landings decreased to 2183 tonnes compared to 3604 tonnes in 2021. In 2023, the official landings decreased even further to 1731 tonnes; 1727 tonnes were landed in 2024. The latter was the lowest official landings value observed for the whole time series.

The main fishing gear in the North Sea is the beam trawl with mesh sizes between 80- and 100-mm. Large effort reductions took place in these fisheries over the last decade (STECF, 2016). The largest part of the landings in Subarea 4 was always taken by the Netherlands, followed by Denmark, the UK, and Belgium (Figure 5.2, Table 5.7). However, in 2023, for the first time, Denmark reported the largest amount of official landings (Subarea 4 and Division 3.a). In Division 3.a, Denmark always landed by far the largest amount of dab (Figure 5.3, Table 5.8). Dab is among the most discarded fish species in the North Sea. In the beam trawl fishery on plaice and sole and the otter trawl fishery on plaice up to 95% of dab catches are discarded (e.g., van Helmond *et al.*, 2012).

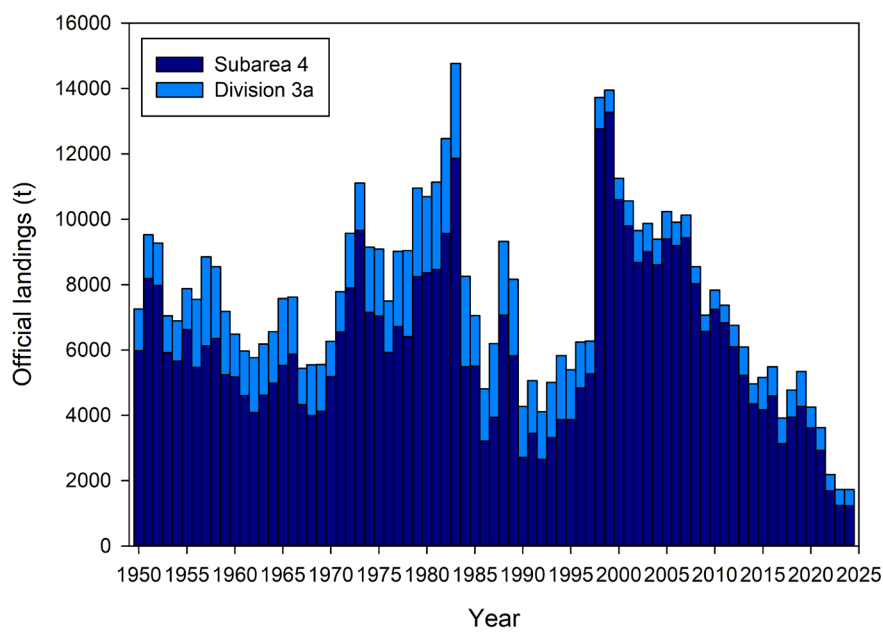


Figure 5.1. Dab in Subarea 4 and Division 3.a. Total official landings of dab in Subarea 4 and Division 3.a in 1950–2024.

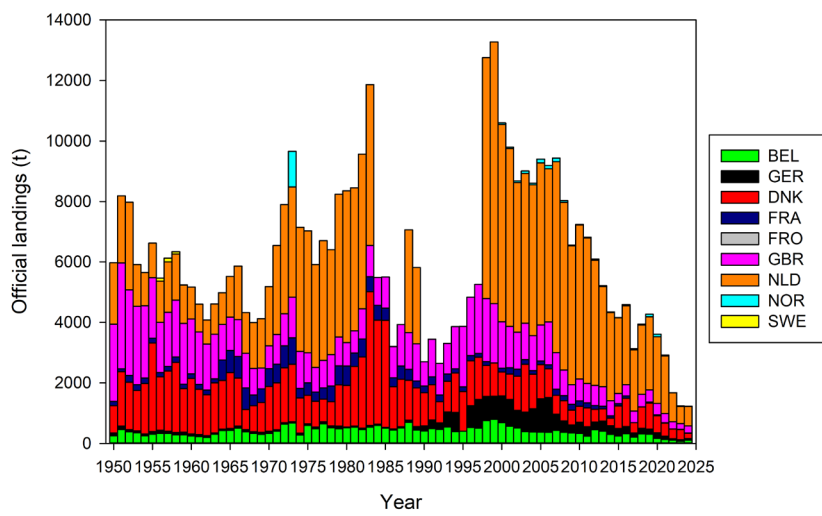


Figure 5.2. Dab in Subarea 4 and Division 3.a. Official landings of dab in Subarea 4 by country 1950 to 2024.

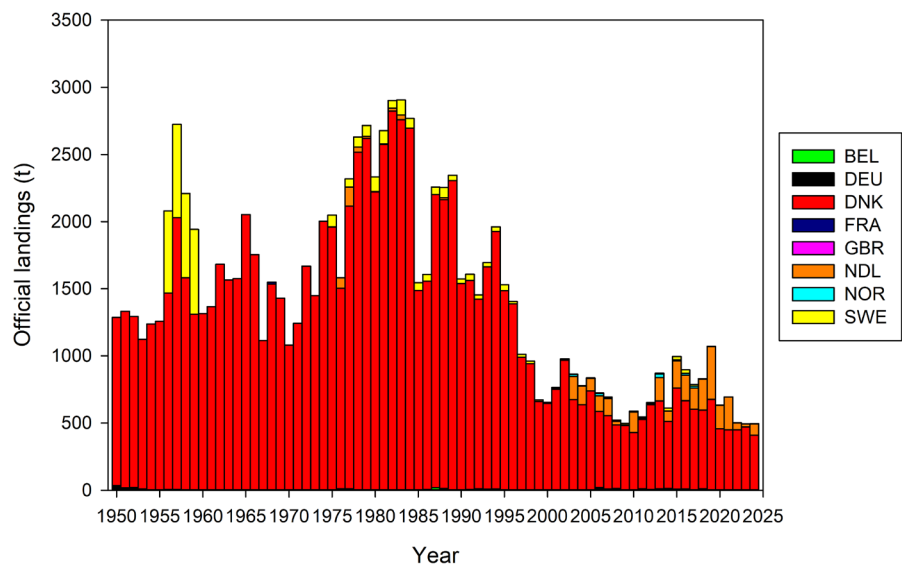


Figure 5.3. Dab in Subarea 4 and Division 3.a. Official landings of dab in Division 3.a by country 1950–2024.

5.2.2 InterCatch

For the current assessment year, dab landing and discard data from 2002–2024 were available in the InterCatch system. Discard information for 2024 was provided for 52% (compared to 60% in 2023, and 62% in 2022) of total landings in relation to weight (Figure 5.4).

In 2025, UK England updated its 2021 to 2023 data. And also the Netherlands updated 2024 InterCatch data. This resulted in changes in total discards and landings for the years 2021 – 2023, listed in the table below (Table 5.1).

Table 5.1. Dab in Subarea 4 and Division 3.a. Changes in discards and landings due to updates in InterCatch. Values in tonnes.

| Year | Total discard (WG 2025) | Total discard (WG 2024) | Difference | Total landings (WG 2025) | Total landings (WG 2024) | Difference |
|------|-------------------------|-------------------------|------------|--------------------------|--------------------------|------------|
| 2021 | 34410                   | 34286                   | -125       | 3321                     | 3321                     | 0          |
| 2022 | 33515                   | 33390                   | -125       | 2171                     | 2169                     | -1         |
| 2023 | 26192                   | 27385                   | 1193       | 1802                     | 1799                     | -3         |

In 2024, the largest catch (landings and discards) was reported by The Netherlands for the TBB\_DEF\_70-99\_0\_0\_all fleet with 10 185 tonnes (Figure 5.5 and Figure 5.6; Table 5.9). The largest amount of landings (262 tonnes) was also reported by the Dutch TBB\_DEF\_70-99\_0\_0\_all fleet. In general, the TBB\_DEF\_70-99\_0\_0\_all fleet landed most dab in the past. But for this fleet the dab landings decreased considerably from 2007 – 2024, with the lowest observed landings in 2024. Other important fleets with > 1000 tonnes of catches are the coastal shrimp fleets (TBB\_CRU\_16-31\_0\_0\_all; 2404 tonnes), *Nephrops* fleets (OTB\_CRU\_70-99\_0\_0\_all; 1628 tonnes) and demersal seiners (SSC\_DEF\_70-99\_0\_0\_all; 1526 tonnes).

By far the largest total catch in 2024 was taken by The Netherlands (15 162 tonnes), followed by Denmark with 2220 tonnes, and Belgium with 1933 tonnes. The total dab catch estimated with InterCatch for 2024 was 23 174 tonnes from which 1716 tonnes were landings and 21 459 tonnes

discards (93% of the total catch, including 5 tonnes BMS landings). It should be noted that not all métiers were sampled in every quarter and that the raising procedure with the InterCatch tool may not be adequate in all cases. Further, there are a number of métiers for which zero landings were reported and a discard raising for these fleets is not possible with the InterCatch tool, which is based on a discard ratio between landings and observed discards. Especially for bycatch species without economic interest, zero landings do not necessarily imply zero discards. However, the Dutch TBB\_DEF\_70-99\_0\_0\_all fleet is by far the most important one in terms of total catch and information on discard weights was provided for every quarter for this fleet.

In general, it was attempted to use the same groupings for discard raising as for the previous data years. However, this was not possible for all cases and compared to the previous year slight changes had to be made. The grouping is generally based on gear type and mesh size and where possible also by area. For the sample allocation scheme landings and discards were grouped by season. The following groupings were used for the 2024 data discard raising:

- Group 1: MIS\_MIS\_HC all area (3.a and 4) raised with all other métiers; no area specific MIS\_MIS\_HC all data were available in 2024 data.
- Group 2: Passive gears area 4 and 3a raised with all passive gears area 4 and 3.a
- Group 3: OTB\_CRU\_70-99\_all raised with OTB\_CRU\_70-99\_all.
- Group 4: OTB\_CRU\_100-119\_0\_0\_all raised with discard rates for all other OTB\_CRU\_100-119\_all métiers (UK ENG only).
- Group 5: OTB\_DEF\_>120\_all area 4 - raised with discard rates available for all other OTB\_DEF\_>120\_all métiers area 4.
- Group 6: OTB\_DEF\_>120\_all area 3.aN - raised with discard rates available for all other OTB\_DEF\_>120\_all métiers area 3.aN.
- Group 7: SSC\_SDN\_DEF>120\_all areas raised with discard rates available for all other SSC\_SDN\_DEF\_>=120\_all métiers.
- Group 8: TBB\_DEF\_70-99 all raised with discard rates available for all other TBB\_DEF\_70-99 all métiers.
- Group 9: TBB\_DEF\_100-119\_all areas raised with discard rates available for all other TBB\_DEF\_100-119 all métiers.
- Group 10: OTB\_DEF\_100-119 raised with discard rates available for all other OTB\_DEF\_100-119 all métiers.
- Group 11: SSC\_DEF\_100-119 (including SSC\_DEF\_all\_all and SDN\_DEF\_all\_all) raised with the OTB\_DEF\_100-119 métier.
- Group 12: OTB\_SSC\_SDN\_DEF\_70-99\_all raised with OTB\_DEF\_70-99\_all and OTB\_CRU\_70-99\_0\_0\_all métiers.
- Group 13: TBB\_DEF\_>=120\_0\_0\_all raised with TBB\_DEF\_>=120\_0\_0\_all metier
- Group 14: OTB\_SSC\_CRU\_\_SPF\_CEP\_DEF\_32-69\_all raised with OTB\_SSC\_CRU\_\_SPF\_CEP\_DEF\_32-69\_all
- Group 15: TBB\_CRU\_16-31\_0\_0\_all raised by all métiers

- No raising for MIS\_MIS\_0\_0\_0\_IBC, OTB\_CRU\_16-31, PS\_SPF\_0\_0\_0.
- Default weighting by CATON
- Length allocation only landings and discard group plus extra discard shrimp group

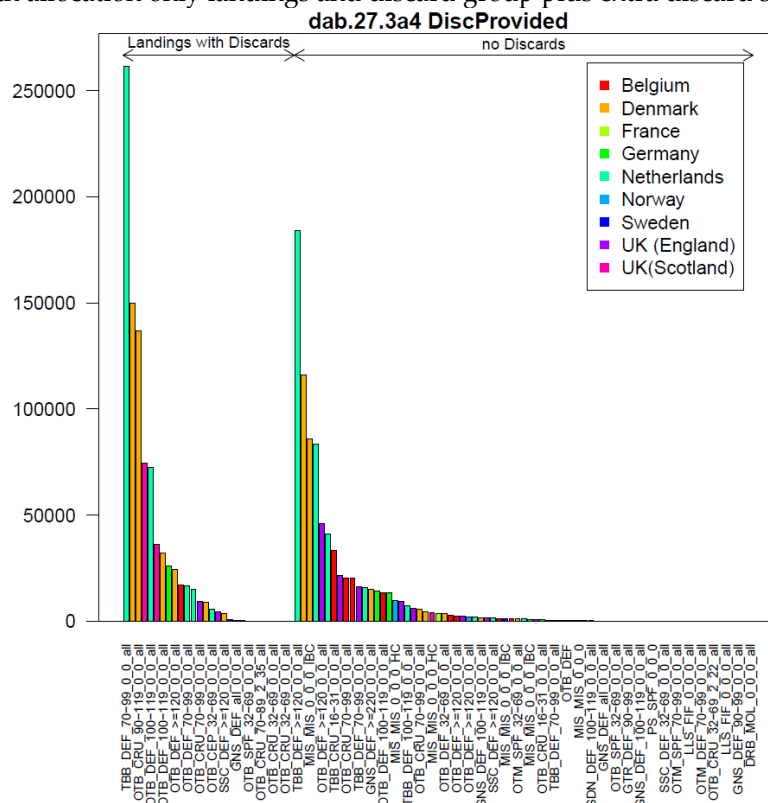


Figure 5.4. Dab in Subarea 4 and Division 3.a. Dab landings and discards (kg) provision for Subarea 4 and Division 3.a by métier and country in 2024 as uploaded to InterCatch.

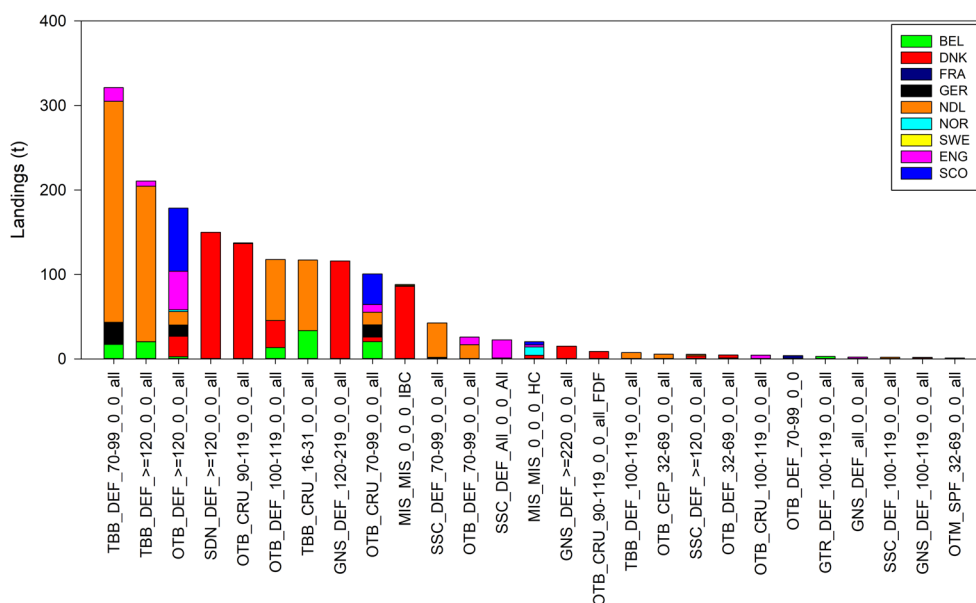


Figure 5.5. Dab in Subarea 4 and Division 3.a. Dab landings (tonnes) for Subarea 4 and Division 3.a by métier and country in 2024 as uploaded to InterCatch.

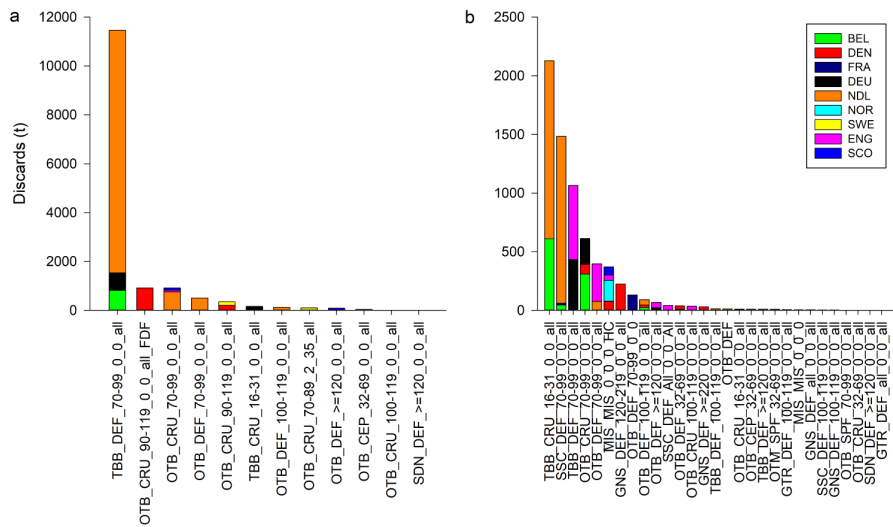


Figure 5.6. Dab in Subarea 4 and Division 3.a. (a) Dab discards for Subarea 4 and Division 3.a by métier and country in 2024. Reported discards (a), raised discards (b).

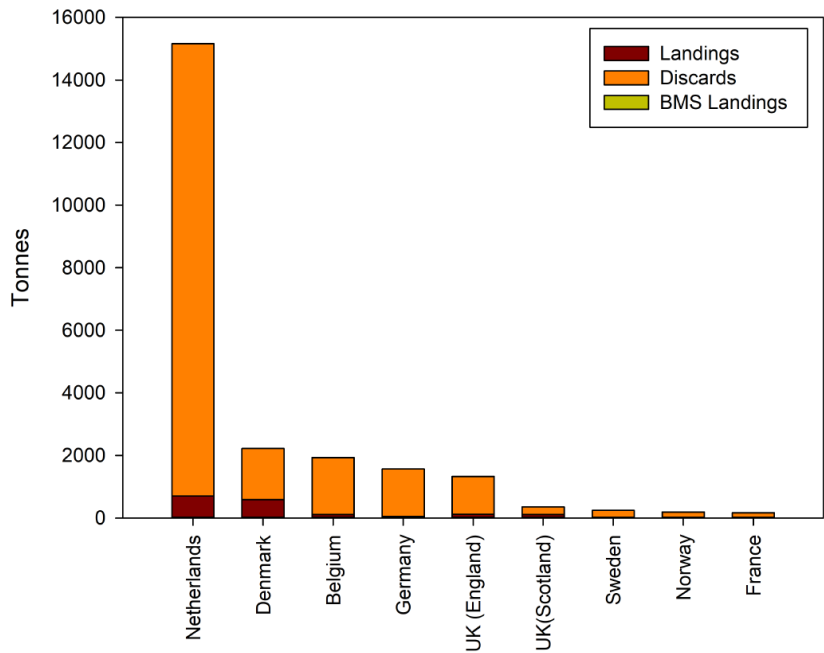


Figure 5.7. Dab in Subarea 4 and Division 3.a: Dab landings and estimated discards for Subarea 4 and Division 3.a by country in 2024.

### 5.3 Survey data/recruit series

Surveys providing information on distribution, abundance and length frequency for dab in Subarea 4 and Division 3.a, are several Beam Trawl Surveys (BTS) in quarter 3 (Figure 5.8 and Figure 5.9) and the International Bottom Trawl Survey (NS-IBTS) in quarter 1 and quarter 3 (Figure 5.10).

The longest beam trawl survey time series exists for the RV Isis covering the southeastern part of the North Sea (Figure 5.9). This index showed some high dab abundance in the earlier years (1987–1990) followed by a sharp decline until 1995. After a second peak in abundance in 1998 the abundance declined again until 2006, and afterwards increased again to such high values as were observed for the time period 1987–1991. The increasing abundance trend from 2005–2006 onwards was also observed for the RV Tridens beam trawl survey, and since 2010 also for the RV Solea beam trawl survey. No clear trend is visible in the RV Belgica survey data. A strong decrease was observed for the RV Solea survey for the year 2015, and again for 2019. Since 2017, RV Isis does not take part any more in the BTS and RV Tridens covers the whole survey area since then. A combined index of the two vessels also displayed a declining trend in general for dab abundance from 2015 onwards. From 2020 onwards, the index was relatively stable, but showed a decrease in the most recent year (2024). Overall, the BTS indices are on a quite stable level since 2016.

The International Bottom Trawl Survey in quarter 1 (NS-IBTS Q1) showed an increasing abundance trend from 1983 to 1990 and fluctuated since then without a clear trend until 2013. From 2013 to 2015 a strong increase in abundance was observed, followed by a strong decrease again in 2017 and 2018 (Figure 5.10). The index decreased in 2024, but increased again in 2025. The NS-IBTS Q3 also showed a highly variable abundance trend with a slight overall increase from the beginning of the time series in 1991 until 2014 (Figure 5.10). Since then, this abundance index steadily decreased until 2019, increased in 2020 and 2021, but decreased again for the recent three years.

In order to estimate a mature biomass index, a length weight relationship and maturity data derived from NS-IBTS Q1 data were estimated. The obtained length weight relationship and the maturity ogive (Figure 5.11) were then applied to estimate the mature biomass index in kg per hour. The mature biomass indices in kg/h (Figure 5.12) show, in general, the same trends as the NS-IBTS abundance indices, and for both quarters the decreasing trend was confirmed until 2018. Since then, the quarter 1 mature biomass index fluctuates without trend, with a higher value for 2025 compared to the previous year. The quarter 3 mature biomass index decreased from 2020 onwards.

Only the beam trawl surveys provide data on age and weight for dab on a regular basis. During the benchmark in 2016, it was agreed to use an age-based survey index combining data from the Dutch and German beam trawl surveys taking into account a possible ship effect (i.e. gear effect; Berg *et al.*, 2014). For age group 1, the index is highly variable and does not show any trends, but a strong decrease from 2017 to 2018 was followed by a strong increase in 2019. For the age groups 2–7, an increase of the index was observed in the most recent year for most of the ages. The indices for the two oldest age groups are variable, but fluctuate on a rather high level compared to early years in the time series. This index served as an input for the survey-based assessment model (SURBAR) to inform the stock status of North Sea dab (Figure 5.13).

The spatial distribution of dab age groups follows a clear pattern with the youngest age groups (0 and 1) located near the coast of the southeastern North Sea and the older age groups more distributed in the central North Sea (Figure 5.14).

The weight at age data shows a slight decreasing trend for all age groups from 2002 to 2015, but an increase since 2016 for the age groups 1–5 (Figure 5.15).



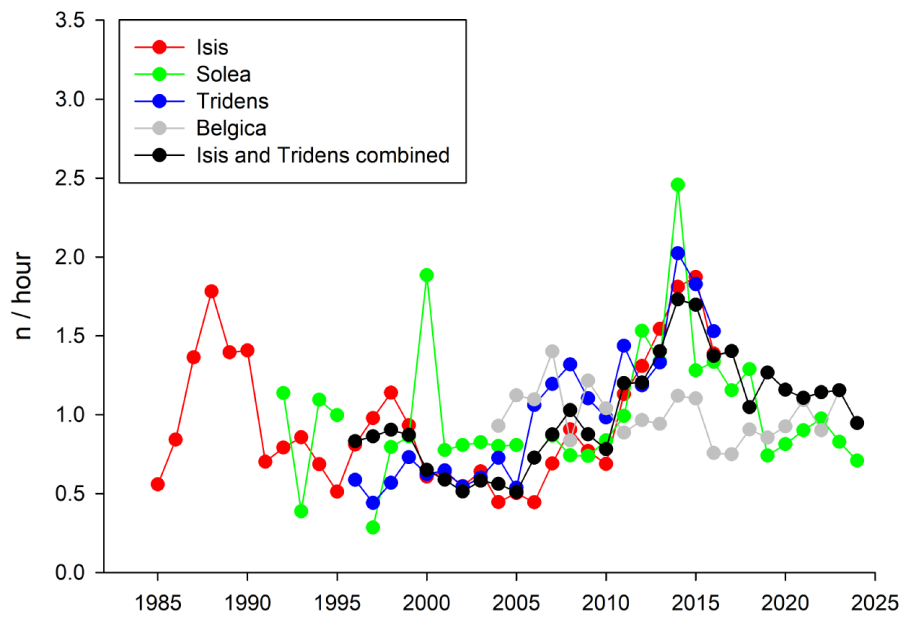


Figure 5.8. Dab in Subarea 4 and Division 3.a. Standardized dab beam trawl survey (BTS) indices (n/hour) in Subarea 4.

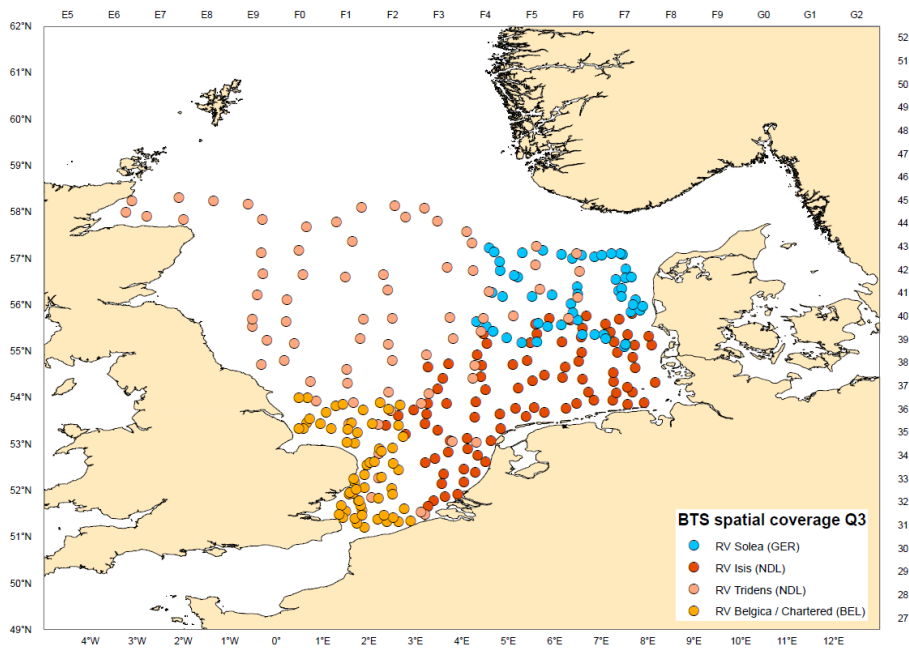


Figure 5.9. Dab in Subarea 4 and Division 3.a. Spatial coverage of the different beam trawl surveys (BTS) in the North Sea. Since 2017, the survey area from RV Isis is also covered by RV Tridens.

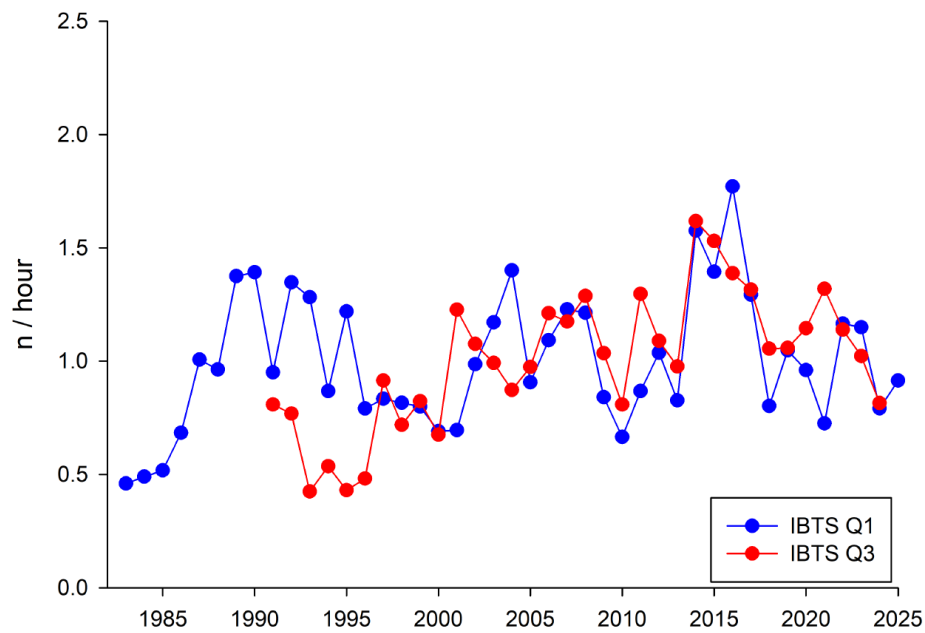


Figure 5.10. Dab in Subarea 4 and Division 3.a. Standardized dab survey indices (n/hour) from the International Bottom Trawl Survey (NS-IBTS).

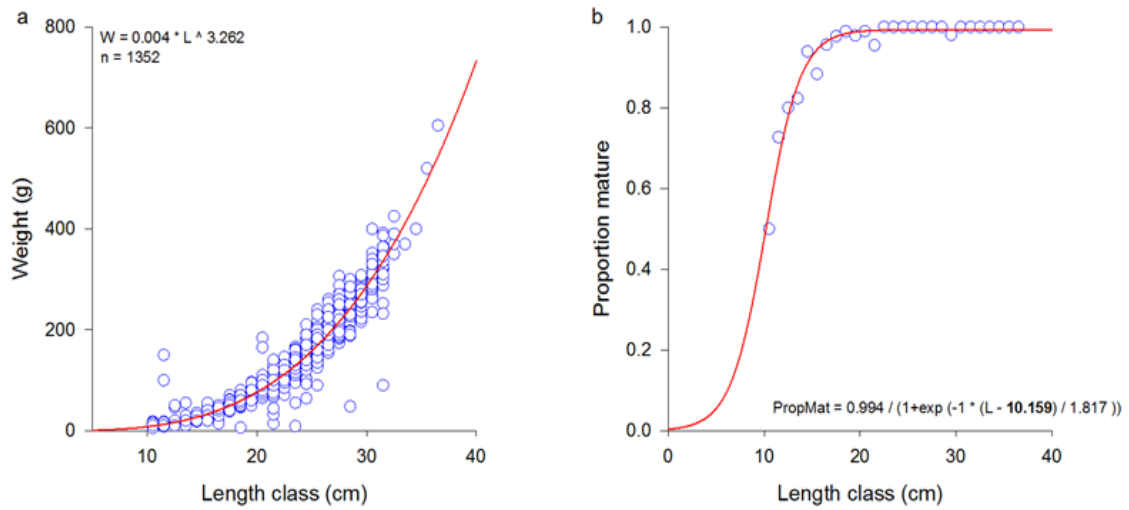


Figure 5.11. Dab in Subarea 4 and Division 3.a. Length weight relation (a) and length-based maturity ogive (b) obtained from survey data (NS-IBTS Q1).

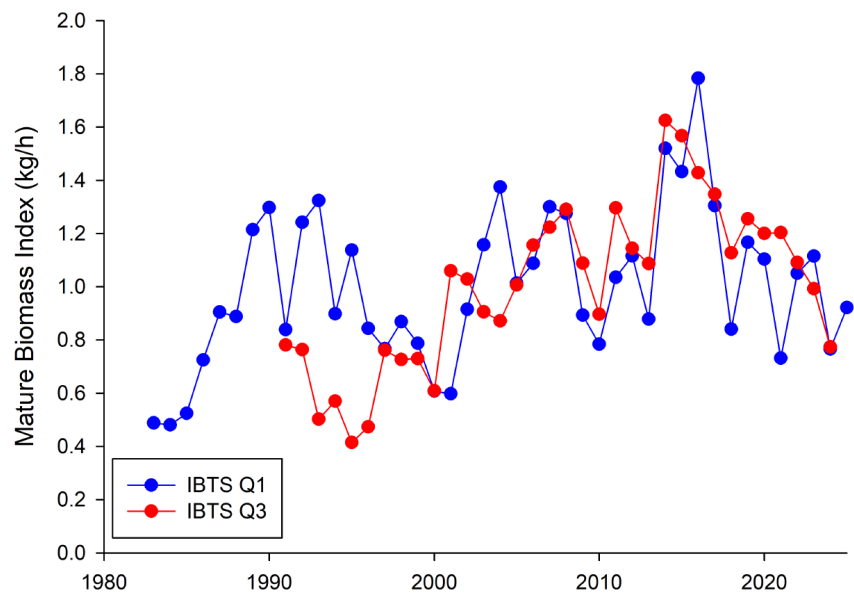


Figure 5.12. Dab in Subarea 4 and Division 3.a. Mature biomass index NS-IBTS Q1 and Q3.

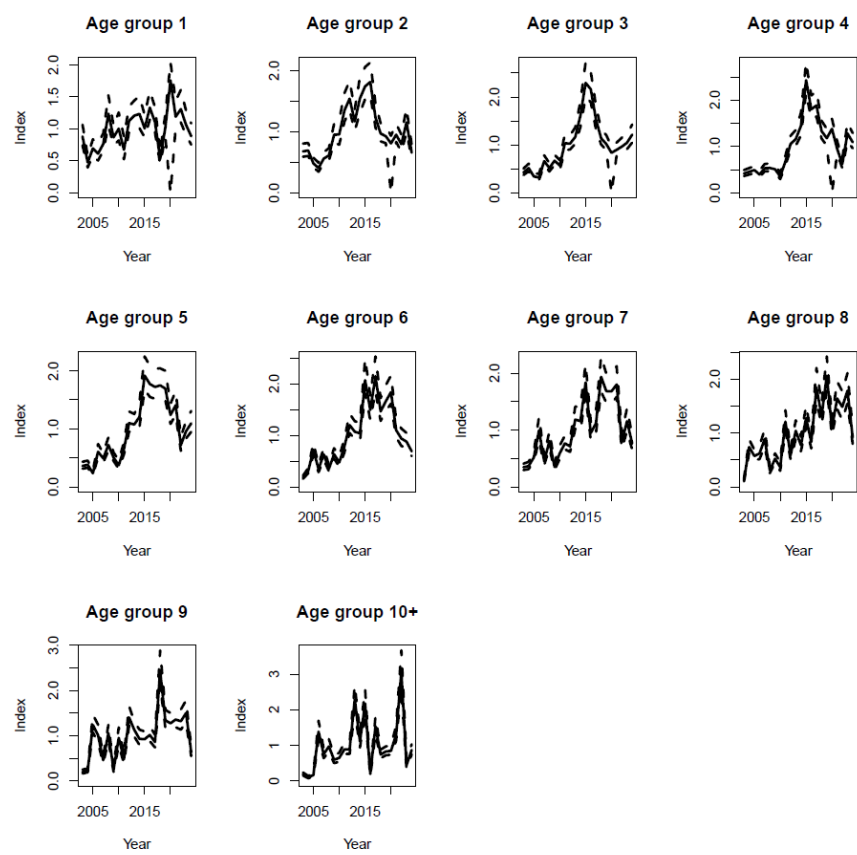


Figure 5.13. Dab in Subarea 4 and Division 3.a. Combined beam trawl index by age groups (2003–2024).

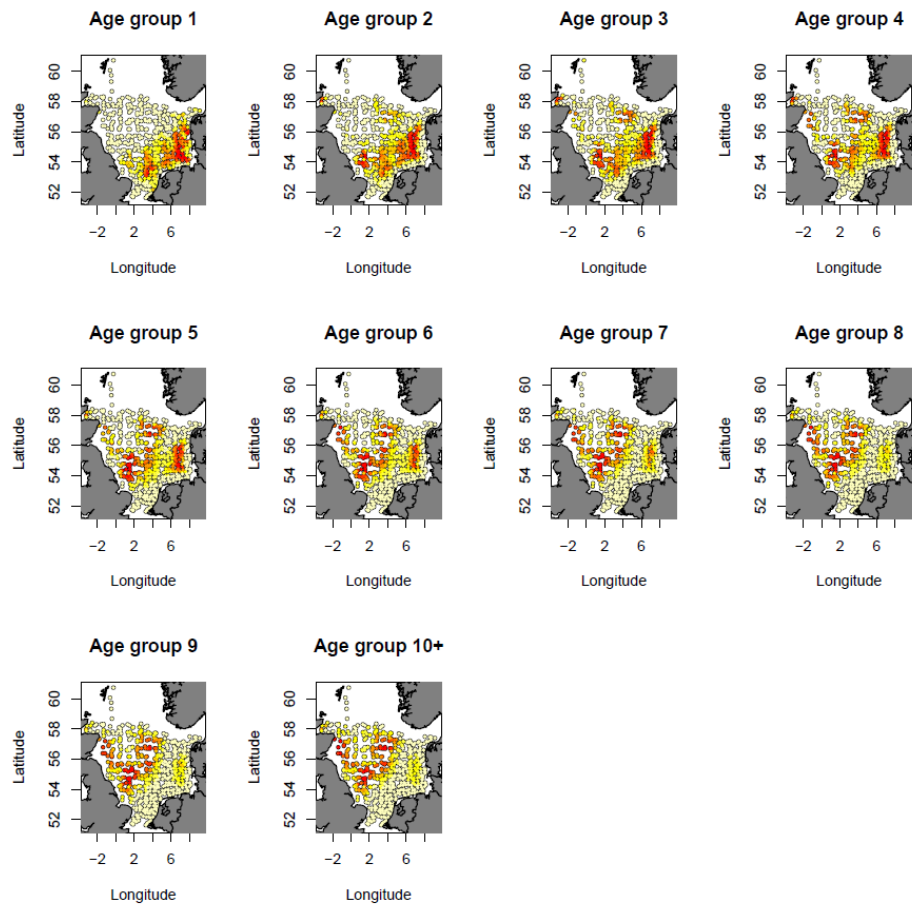


Figure 5.14. Dab in Subarea 4 and Division 3.a. Dab distribution in the North Sea by age group obtained by the Dutch and German Beam Trawl Surveys (1985–2024).

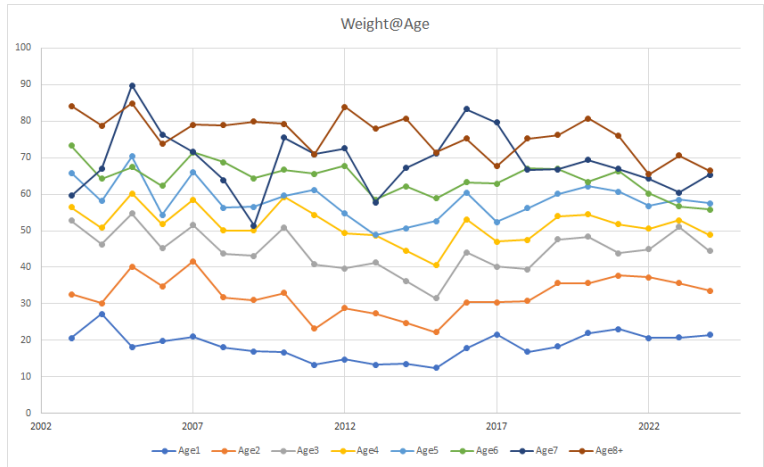


Figure 5.15. Dab in Subarea 4 and Division 3.a. Weight at age derived from beam trawl survey data (2003–2024).

### 5.4 Survey Based Assessment (SURBAR)

In 2016, a benchmark assessment was carried out for dab (ICES, 2016). During this benchmark it was agreed to make use of the available data from the beam trawl surveys and to run a survey-based assessment model (SURBAR; Needle, 2015), taking the age structure of dab into account. The SURBAR results of the update assessment showed no clear trend in total mortality ( $z$ ) for the years 2003–2024 (Figure 5.16, upper left panel). The mortality was similar compared to last year’s value. The spawning stock biomass (relative biomass) increased for the years 2003–2016 (Figure 5.16, upper right panel), but decreased after the peak in 2016 until 2022. The 2023 value showed an increase again, but another decrease for 2024. The total stock biomass follows the trend of the SSB. The recruitment increased by a factor of 2.6 from 2003 to 2014, but decreased since 2015 until 2022 (Figure 5.16, lower right panel) and displayed a stronger decrease for 2024 again. No pattern was detected in the log residual pattern of the age-based survey indices (Figure 5.17). The effects of age, year and cohorts on the SURBAR model are displayed in Figure 5.18. Catch curves overall look reasonable (Figure 5.19). However, for some catch curves an upward trend is visible, e.g. for the 2005-2007 cohorts. The standardized SURBAR indices by age group (Figure 5.20) are in line with the overall index trend. There was a strong pattern in the retrospective for total mortality observed, but still within the confidence limits (Figure 5.21). The retrospective analyses for TSB, SSB and  $r$  do not show strong patterns. The SURBAR results are summarized in Table 5.2.

Table 5.2. Dab in Subarea 4 and Division 3.a. Settings and input data used for the final SURBAR assessment run.

| Setting/Data   | Values/source                                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Survey index   | Combined beam trawl survey index 2003–current assessment year (BTS-Isis, BTS-Tridens, German BTS). Delta GAM Method by Berg <i>et al.</i> (2014). |
| Ages           | 1–6                                                                                                                                               |
| Lambda         | 3                                                                                                                                                 |
| zbar           | 1–6                                                                                                                                               |
| Spawning time  | 0.4                                                                                                                                               |
| Maturity ogive | Fixed ogive, age 1 = 60%, age 2 = 80%, age 3 and older 100%                                                                                       |
| Weight at age  | Data from Dutch Beam Trawl Surveys (2003–current assessment year)                                                                                 |

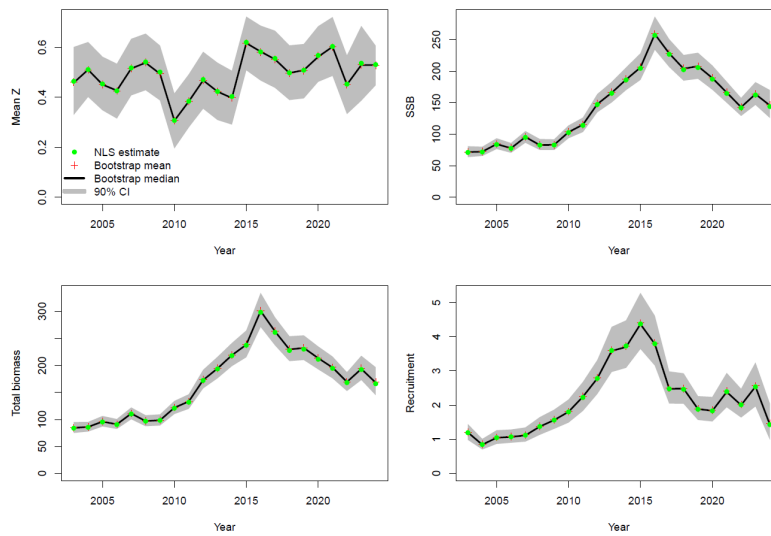


Figure 5.16. Dab in Subarea 4 and Division 3.a. SURBAR model results for dab total mortality ( $z$ ), spawning stock biomass (SSB), total stock biomass (TSB) and recruitment.



Figure 5.17. Dab in Subarea 4 and Division 3.a. SURBAR model results of log residuals.

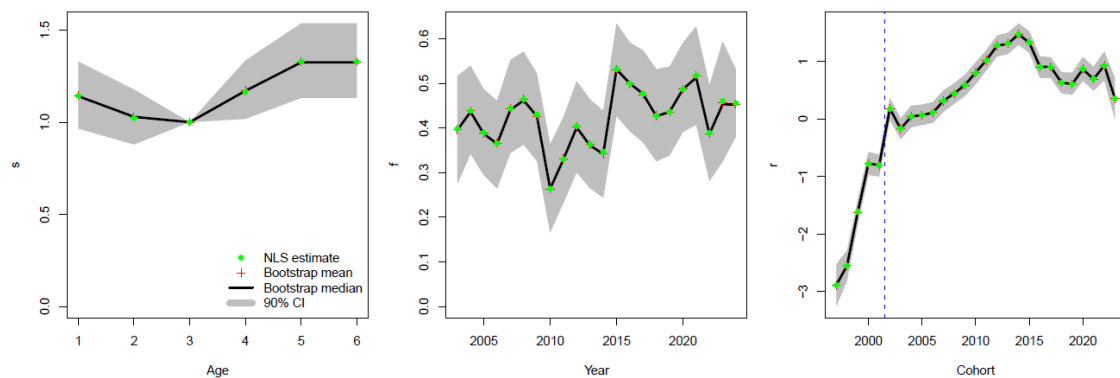


Figure 5.18. Dab in Subarea 4 and Division 3.a. SURBAR model results displaying the age, year and cohort effects.

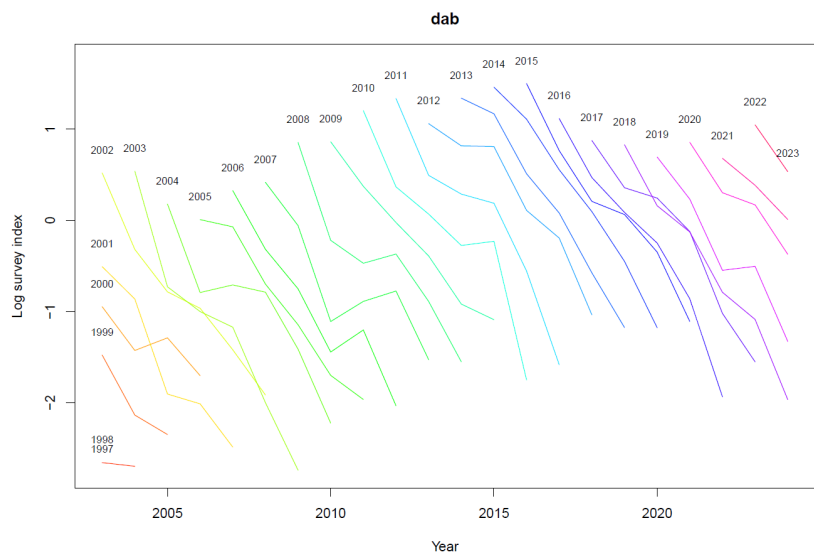


Figure 5.19. Dab in Subarea 4 and Division 3.a. SURBAR model results: catch curves.



Figure 5.20. Dab in Subarea 4 and Division 3.a. SURBAR mean-standardized log survey index by age classes (1-6yr).

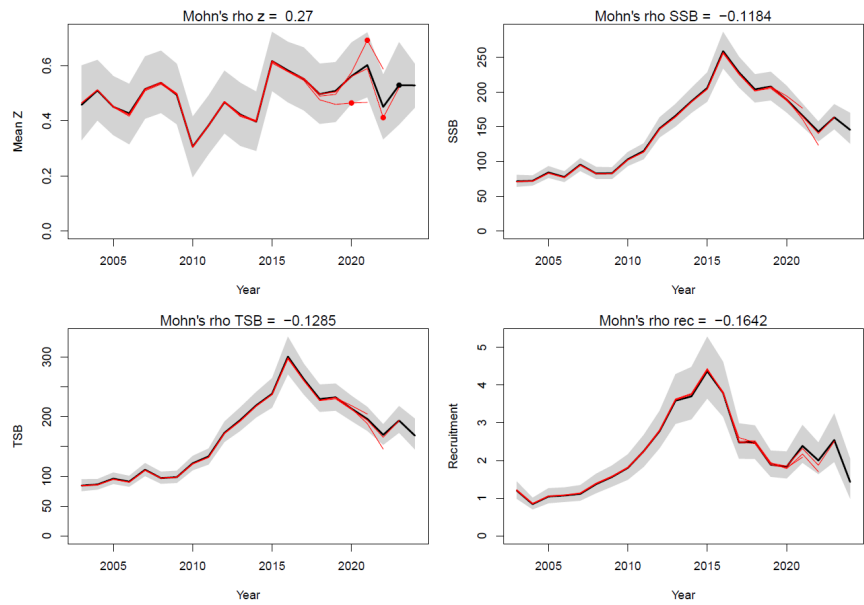


Figure 5.21. Dab in Subarea 4 and Division 3.a. SURBAR Retrospective runs with corresponding Mohn’s rho values.

5.5 MSY proxy analyses for dab in Subarea 4 and Division 3.a

5.5.1 Surplus Production Model in Continuous Time (SPiCT)

In order to estimate MSY proxy reference points for dab a Surplus Production Model in Continuous Time (SPiCT; Pedersen and Berg, 2017) was applied. Three fishery independent survey time series and a catch time-series (InterCatch, 2002–2024) were used as input for the model (details of model input and settings given in Table 5.3). The survey time series were reduced by the recruits (i.e. > 12 cm or > age 1) in order to obtain a better proxy for the exploitable biomass, which is a prerequisite for any production model.

Table 5.3. Dab in Subarea 4 and Division 3.a. SPiCT settings and input data.

| Setting/Data                        | Values/Source             |
|-------------------------------------|---------------------------|
| Catch time series                   | InterCatch data 2002–2024 |
| BTS Isis                            | 1987–2002, >12 cm         |
| BTS Tridens                         | 1996–2002, >12 cm         |
| Combined BTS (Isis, Tridens, Solea) | 2003–2024, Age > 1 yr     |
| SPiCT settings                      | no priors                 |

The results of the SPiCT assessment for dab in Subarea 4 and Division 3.a (Figures 5.22 – 5.27) showed that the relative fishing mortality is below the reference  $F_{MSY}$  proxy (Figure 5.23, right panel) and the relative biomass is above the reference  $B_{MSY} * 0.5$  proxy (Figure 5.24; left panel). Moreover, the estimated uncertainty boundaries around the relative F values show that these are below the reference  $F_{MSY}$  proxy for recent years, and those estimated for the relative biomass are above the reference  $B_{MSY} * 0.5$  for recent years. However, it must be noted that the absolute F and



biomass estimates are highly uncertain and must not be used for any further analyses or conclusions. The SPiCT model was rejected by the working group in 2022, and it was not considered as the basis of the catch advice since then. However, it could be reconsidered during a benchmark assessment for this stock in the future.

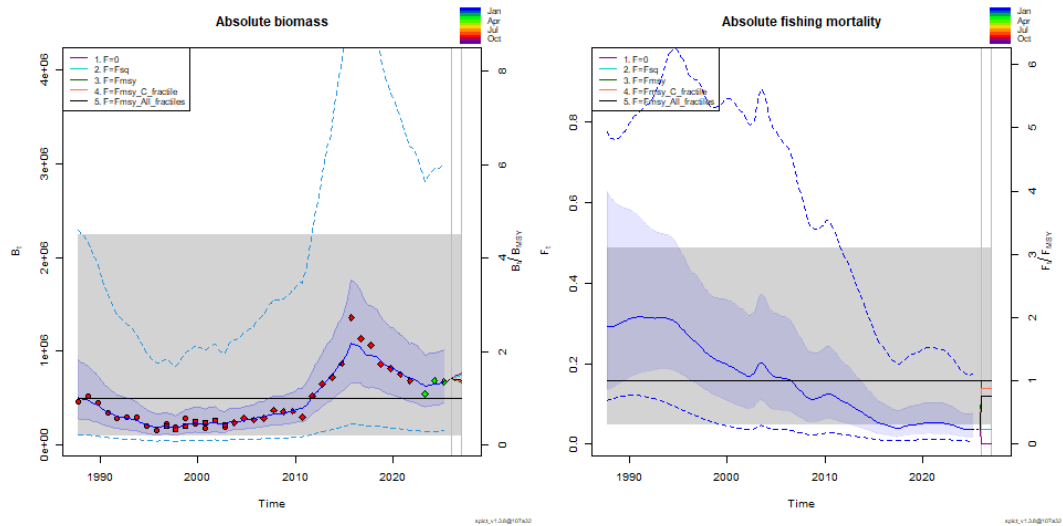


Figure 5.22. Dab in Subarea 4 and Division 3.a. SPiCT results. Absolute biomass (left panel) and absolute fishing mortality (right panel).

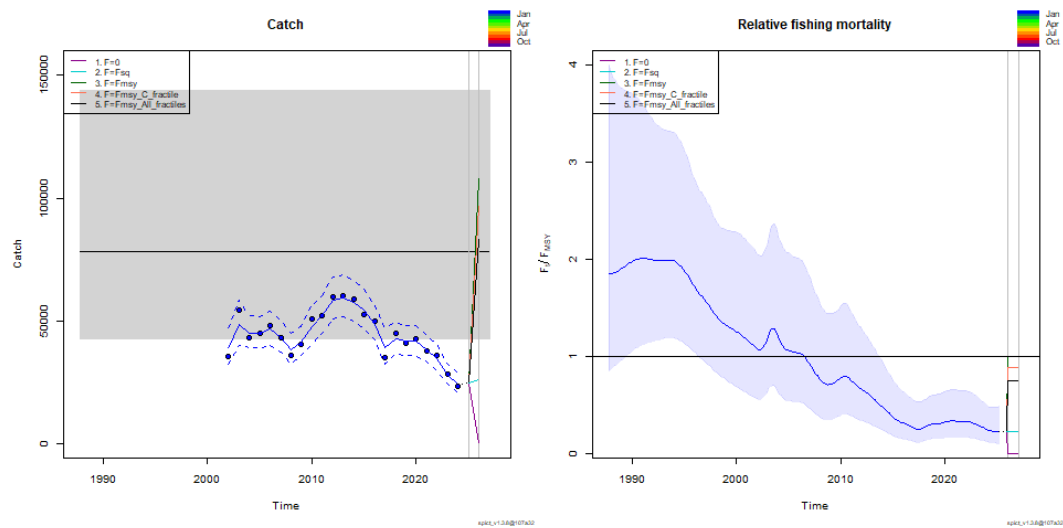


Figure 5.23. Dab in Subarea 4 and Division 3.a. SPiCT results. Catch time series (left panel) and relative fishing mortality (right panel).

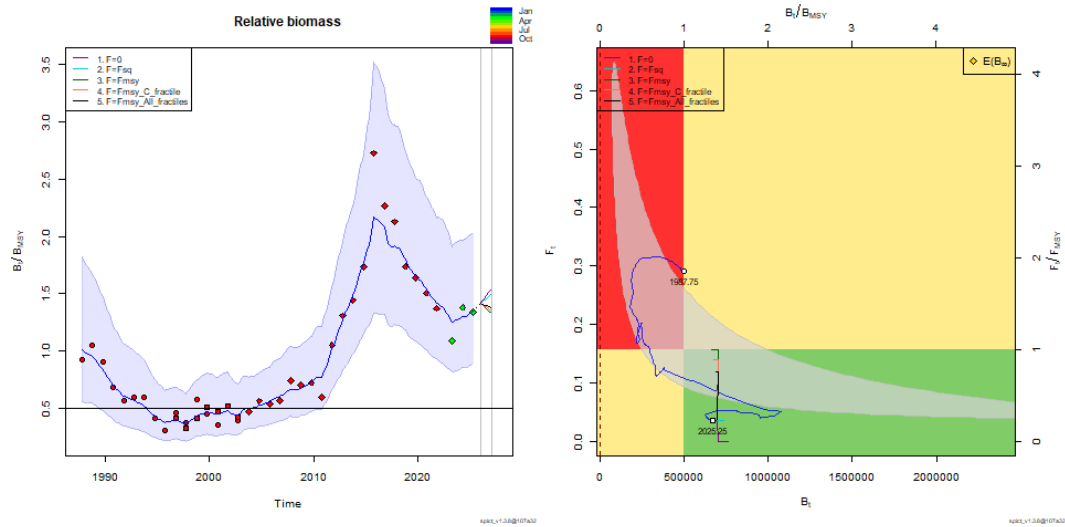


Figure 5.24. Dab in Subarea 4 and Division 3.a. SPiCT results. Relative biomass (left panel) and Kobe plot of relative fishing mortality over biomass estimate (right panel).

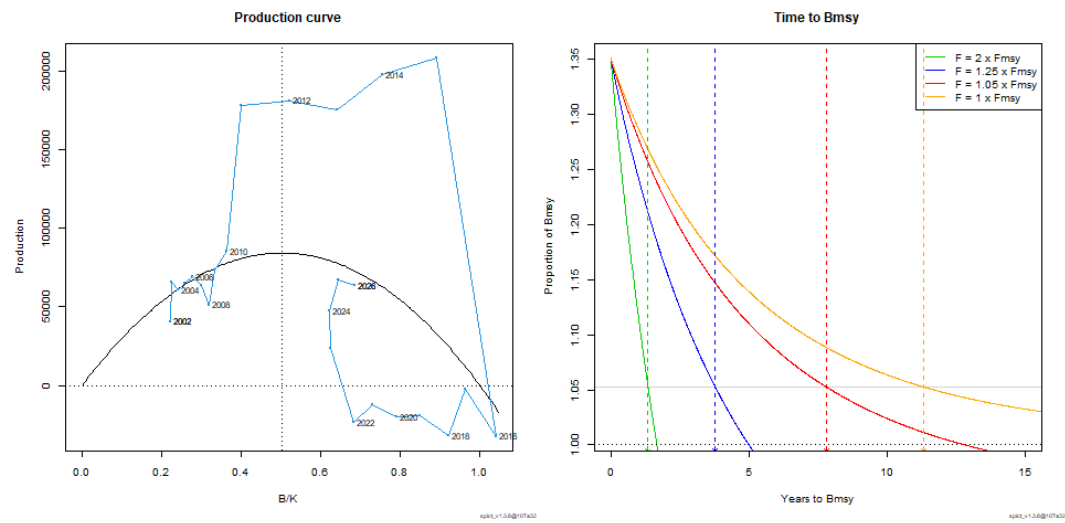


Figure 5.25. Dab in Subarea 4 and Division 3.a. SPiCT results. Production curve (left panel) and estimated time to  $B_{MSY}$  (right panel).

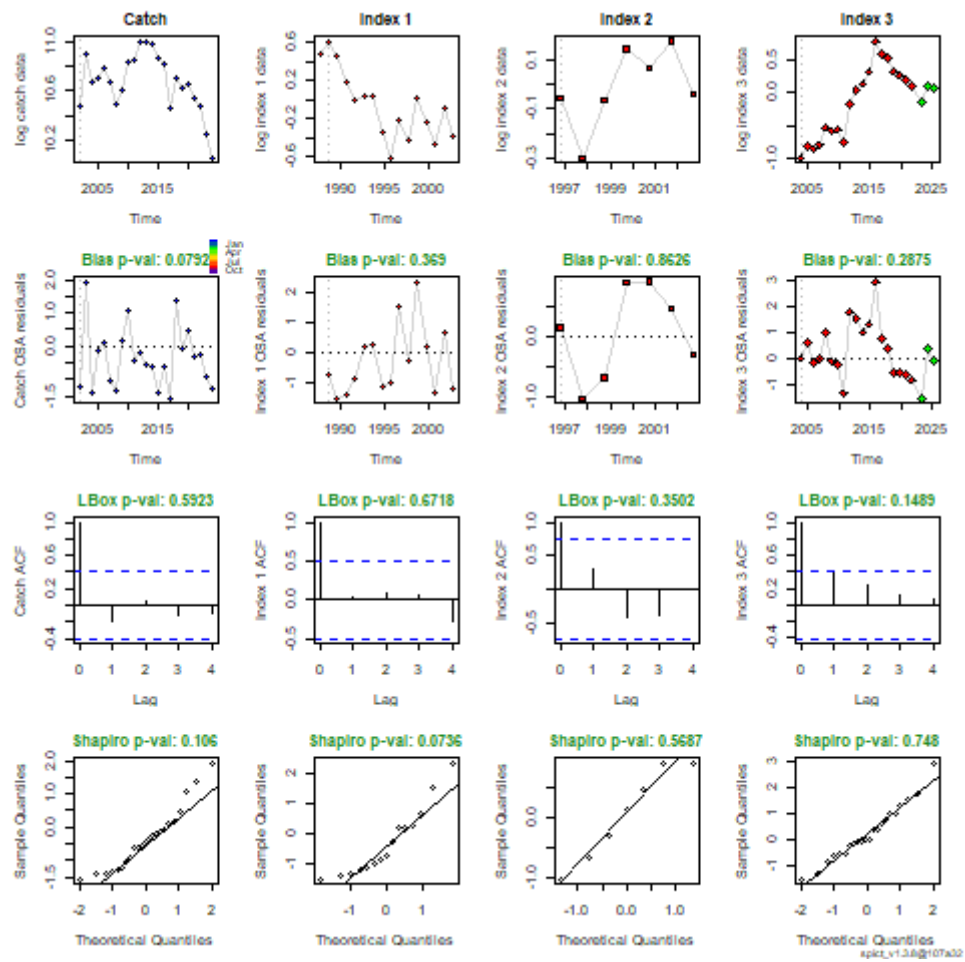


Figure 5.26. Dab in Subarea 4 and Division 3.a. SPIC diagnostics.

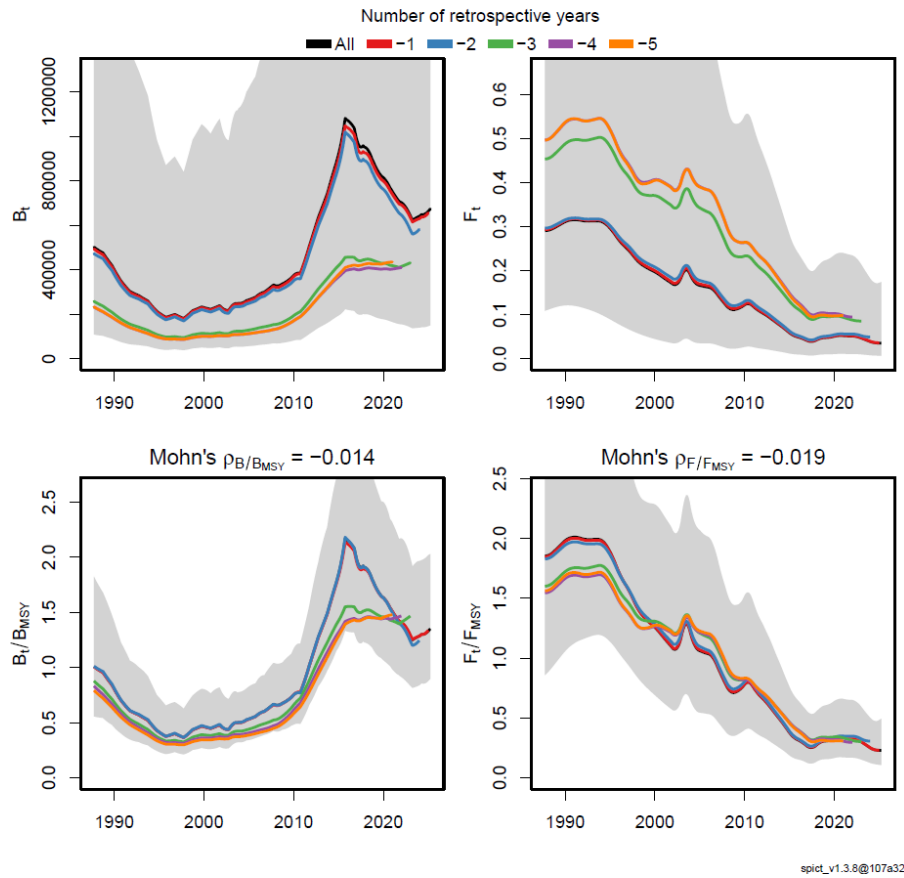


Figure 5.27. Dab in Subarea 4 and Division 3.a. SPiCT retrospective plots.

### 5.5.2 Length Based Indicator Method (LBI)

The Length Based Indicator method was applied in order to evaluate the stock status relative to MSY proxies. Length data from commercial samplings uploaded to InterCatch were used as input (2014–2024). During the last dab benchmark assessment in 2016 (ICES, 2016), von Bertalanffy growth parameters were estimated from survey data ( $L_{inf} = 24.9$  cm,  $k = 0.39$ ,  $t_0 = -0.7$ ; BTS data 2003 - 2015). The  $L_{mat50\%}$  was derived from literature (13.0 cm; Froese and Sampang, 2013). For the analysis the data were corrected for recruits, i.e. only lengths  $>10.5$  cm were taken into account. Individuals below this size are mainly caught in high numbers by the shrimper fleet near the coast and appear in the discard data. The resulting length distributions are very similar from year to year (Figure 5.28). With respect to conservation, all indicator ratios are above the reference (Table 5.4). This is also the case for all other indicators. The  $L_{mean} / L_{F=M}$  ratio indicates that  $F$  was below  $F_{MSY proxy}$  for this stock for the whole time period 2014–2024.

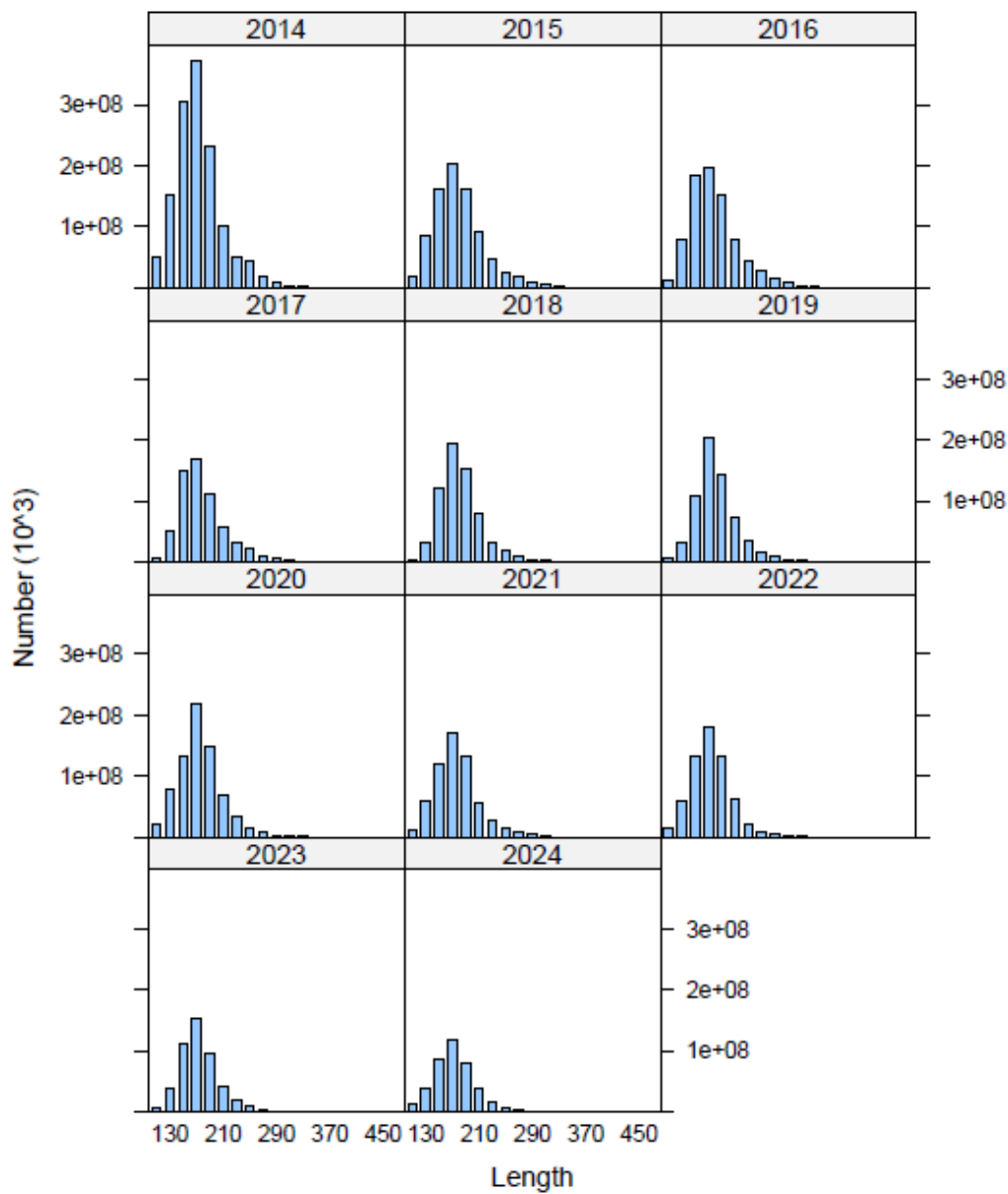


Figure 5.28 Dab in Subarea 4 and Division 3.a. Length distributions from commercial samplings. InterCatch data 2014 – 2024.

Table 5.4 Dab in Subarea 4 and Division 3.a. Summary of the Length Based Indicator Method.

| Ref  | Conservation  |                    |                      | Optimizing Yield |                    | MSY                |
|------|---------------|--------------------|----------------------|------------------|--------------------|--------------------|
|      | $L_C/L_{mat}$ | $L_{25\%}/L_{mat}$ | $L_{max5\%}/L_{inf}$ | $P_{mega}$       | $L_{mean}/L_{opt}$ | $L_{mean}/L_{F=M}$ |
| 2014 | >1            | >1                 | >0.8                 | >30%             | ~1(>0.9)           | ≥1                 |
| 2015 | 1.15          | 1.12               | 1.07                 | 0.35             | 1.12               | 1.07               |
| 2016 | 1.15          | 1.19               | 1.11                 | 0.43             | 1.15               | 1.09               |
| 2017 | 1.15          | 1.19               | 1.10                 | 0.41             | 1.14               | 1.09               |
| 2018 | 1.15          | 1.19               | 1.11                 | 0.39             | 1.13               | 1.08               |
| 2019 | 1.15          | 1.27               | 1.08                 | 0.46             | 1.13               | 1.08               |
| 2020 | 1.15          | 1.27               | 1.08                 | 0.44             | 1.13               | 1.07               |
| 2021 | 1.15          | 1.19               | 1.04                 | 0.38             | 1.12               | 1.06               |
| 2022 | 1.15          | 1.19               | 1.07                 | 0.41             | 1.13               | 1.07               |
| 2023 | 1.15          | 1.19               | 1.00                 | 0.38             | 1.11               | 1.05               |
| 2024 | 1.15          | 1.19               | 1.02                 | 0.36             | 1.10               | 1.05               |
| 2024 | 1.15          | 1.19               | 1.02                 | 0.36             | 1.11               | 1.05               |

## 5.6 Empirical *chr* rule

The growth parameter  $k = 0.39$  was estimated during the last benchmark assessment for dab (ICES, 2016). Therefore, the constant harvest rate (*chr*) rule (ICES, 2020) was applied in the case of dab.

The *chr* rule applies a constant harvest rate that is considered a proxy for an MSY harvest rate, which is then applied to a biomass index:

$$C_{y+1} = I_{y-1} \times F_{MSYproxy} \times b \times m$$

with  $C_{y+1}$  is the catch advice,  $I_{y-1}$  the most recent index value,  $F_{MSYproxy}$  is the ratio  $C_y / I_y$  for which the quantity  $f > 1$ , where  $f$  is the ratio of the mean length  $L_{mean}$  in the catch and  $L_{F=M}$ ,  $b$  is a biomass safeguard, and  $m$  is a multiplier applied to the *chr* rule to maintain the probability of the biomass declining below  $B_{lim}$  to less than 5%.

The index used as basis for the *chr* calculation was the biomass index from the SURBAR assessment (Table 5.10).

The biomass safeguard  $b$  is determined by the relation of the most recent index value to  $I_{trigger}$  (with  $I_{trigger} = 1.4 \times I_{loss}$ ;  $I_{loss}$  = defined as lowest observed index value). When the most recent index data is greater than  $I_{trigger}$ ,  $b$  is set to 1. This was the case for dab. The precautionary multiplier  $m$  is applied with the *chr* rule to maintain the probability of the biomass declining below  $B_{lim}$  to less than 5%. Following the guidelines (ICES, 2022), this precautionary multiplier should be set to 0.5 when applying the *chr* rule. An uncertainty cap (*stability clause*) is applied, which limits the amount the advised catch can change upwards or downwards between years (+20% / -30%). This uncertainty cap was applied in the case of dab. The new catch advice was given relative to the most recent catch advice for the years 2023 – 2025 (no more than 28 238 tonnes).

Applying the *chr* rule to the dab stock resulted in a catch advice of no more than 16 163 tonnes which is a change in advice of -43%. Therefore, the stability clause was applied and the new catch advice was 19 767 tonnes (-30%). This corresponds to 1 313 tonnes landings, if the average discard rate of 93% does not change (Table 5.5). The change in advice is due to the decrease in the biomass index and the subsequent application of the stability clause.

**Table 5.5 Dab in Subarea 4 and Division 3.a. The basis for the catch scenarios. Catches are in tonnes. \***

|                                                                                                                                                          |               |      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|
| Previous catch advice $A_y$ (2023, 2024, and 2025)                                                                                                       | 28 238 tonnes |      |
| <b>Biomass index</b>                                                                                                                                     |               |      |
| I: most recent biomass index ( $I_{2024}$ )                                                                                                              | 143.855       |      |
| <b>MSY proxy harvest rate</b>                                                                                                                            |               |      |
| $F_{MSY\ proxy}$ : MSY proxy harvest rate (average of the ratio of catch to biomass index for the years for which $f>1$ , where $f = L_{mean}/L_{F=M}$ ) | 224.718       |      |
| <b>Biomass safeguard</b>                                                                                                                                 |               |      |
| $I_{2003} = I_{loss}$                                                                                                                                    | 71.097        |      |
| Index trigger value ( $I_{trigger}=I_{loss}\times1.4$ )                                                                                                  | 99.536        |      |
| Index 2024 ( $I_{2024}$ )                                                                                                                                | 143.855       |      |
| $b$ : index relative to trigger value, $\min\{I_{2024}/I_{trigger}, 1\}$                                                                                 | 1.00          |      |
| <b>Precautionary multiplier to maintain biomass above <math>B_{lim}</math> with 95% probability</b>                                                      |               |      |
| $m$ : multiplier (generic multiplier based on life history)                                                                                              | 0.5           |      |
| Stability clause (+20%/-30% compared to $A_y$ , only considered if $b=1$ )                                                                               | Applied       | -30% |
| Discard rate                                                                                                                                             | 93%           |      |
| Catch advice for 2026, 2027, 2028**                                                                                                                      | 19 767 tonnes |      |
| % advice change***                                                                                                                                       | -30 %         |      |
| Corresponding landings                                                                                                                                   | 1 313 tonnes  |      |

\* The figures in the table are rounded. Calculations were done with unrounded inputs, and computed values may not match exactly when calculated using the rounded figures in the table.

\*\* Formula:  $I \times F_{MSYproxy} \times b \times m$ , limited by stability clause if applicable.

\*\*\* Advice value for 2026, 2027, 2028 relative to the most recent catch advice (28 238 tonnes).

## 5.7 Issues list

- Métiers with zero landings but no discards reported: No raising possible for these cases. What is the possible impact on catch estimation? Are there other ways to estimate realistic discards for these métiers?
- No suitable data available for the shrimper fleets operating in coastal waters. No reliable discard raising possible for these fleets. What is the possible impact on catch estimation? Is there another way to estimate the discards of these fleets?
- Investigate extending the delta-GAM index with Belgian and German BTS data (prior to 2002). Further, the delta-GAM index still uses the haul duration to standardize number at length. However, the Working Group on Beam Trawl Surveys (ICES, 2021b) recommends to use swept area as standardization. This should be changed and the currently used index and the survey trends based on swept area should be compared.
- Investigate the use of DYFS, DFS inshore surveys to estimate a recruitment index.
- SPiCT was no longer accepted by the WGNSSK; possible improvements should be investigated to obtain an acceptable SPiCT model, which could then be used as basis for advice.
- Investigate which effort data are available and if these could be used as further input for the SPiCT model.

## 5.8 References

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## 5.9 Tables

**Table 5.6. Dab in Subarea 4 and Division 3.a. Official dab landings by ICES area from 1950 – 2024.**

| Year | Subarea 4 | Division 3.a | Total |
|------|-----------|--------------|-------|
| 1950 | 5971      | 1287         | 7258  |
| 1951 | 8190      | 1332         | 9522  |
| 1952 | 7976      | 1294         | 9270  |
| 1953 | 5915      | 1123         | 7038  |
| 1954 | 5652      | 1237         | 6889  |
| 1955 | 6623      | 1257         | 7880  |
| 1956 | 5468      | 2081         | 7549  |
| 1957 | 6127      | 2724         | 8851  |
| 1958 | 6342      | 2210         | 8552  |
| 1959 | 5239      | 1943         | 7182  |
| 1960 | 5168      | 1314         | 6482  |
| 1961 | 4602      | 1367         | 5969  |
| 1962 | 4082      | 1683         | 5765  |



| Year | Subarea 4 | Division 3.a | Total |
|------|-----------|--------------|-------|
| 1963 | 4615      | 1565         | 6180  |
| 1964 | 4982      | 1575         | 6557  |
| 1965 | 5519      | 2052         | 7571  |
| 1966 | 5862      | 1755         | 7617  |
| 1967 | 4324      | 1115         | 5439  |
| 1968 | 3995      | 1548         | 5543  |
| 1969 | 4122      | 1430         | 5552  |
| 1970 | 5183      | 1079         | 6262  |
| 1971 | 6546      | 1242         | 7788  |
| 1972 | 7901      | 1669         | 9570  |
| 1973 | 9657      | 1449         | 11106 |
| 1974 | 7146      | 2003         | 9149  |
| 1975 | 7033      | 2049         | 9082  |
| 1976 | 5917      | 1583         | 7500  |
| 1977 | 6702      | 2318         | 9020  |
| 1978 | 6407      | 2630         | 9037  |
| 1979 | 8243      | 2716         | 10959 |
| 1980 | 8357      | 2333         | 10690 |
| 1981 | 8454      | 2679         | 11133 |
| 1982 | 9565      | 2902         | 12467 |
| 1983 | 11865     | 2906         | 14771 |
| 1984 | 5482      | 2769         | 8251  |
| 1985 | 5502      | 1545         | 7047  |
| 1986 | 3205      | 1608         | 4813  |
| 1987 | 3931      | 2258         | 6189  |
| 1988 | 7067      | 2254         | 9321  |
| 1989 | 5816      | 2346         | 8162  |
| 1990 | 2701      | 1574         | 4275  |
| 1991 | 3448      | 1609         | 5057  |

| Year | Subarea 4 | Division 3.a | Total |
|------|-----------|--------------|-------|
| 1992 | 2647      | 1454         | 4101  |
| 1993 | 3309      | 1695         | 5004  |
| 1994 | 3861      | 1961         | 5822  |
| 1995 | 3865      | 1530         | 5395  |
| 1996 | 4834      | 1405         | 6239  |
| 1997 | 5259      | 1012         | 6271  |
| 1998 | 12759     | 961          | 13720 |
| 1999 | 13276     | 673          | 13949 |
| 2000 | 10595     | 654          | 11249 |
| 2001 | 9799      | 765          | 10564 |
| 2002 | 8678      | 977          | 9655  |
| 2003 | 9008      | 865          | 9873  |
| 2004 | 8608      | 779          | 9387  |
| 2005 | 9402      | 836          | 10238 |
| 2006 | 9189      | 725          | 9914  |
| 2007 | 9433      | 694          | 10127 |
| 2008 | 8029      | 522          | 8551  |
| 2009 | 6562      | 498          | 7060  |
| 2010 | 7241      | 589          | 7830  |
| 2011 | 6827      | 545          | 7372  |
| 2012 | 6096      | 653          | 6749  |
| 2013 | 5213      | 871          | 6084  |
| 2014 | 4346      | 611          | 4957  |
| 2015 | 4162      | 995          | 5157  |
| 2016 | 4590      | 895          | 5485  |
| 2017 | 3130      | 788          | 3918  |
| 2018 | 3940      | 830          | 4770  |
| 2019 | 4273      | 1069         | 5342  |
| 2020 | 3615      | 637          | 4252  |

| Year  | Subarea 4 | Division 3.a | Total |
|-------|-----------|--------------|-------|
| 2021  | 2924      | 696          | 3620  |
| 2022  | 1681      | 502          | 2183  |
| 2023* | 1237      | 494          | 1731  |
| 2024* | 1231      | 496          | 1727  |

\* Preliminary catch statistics

**Table 5.7. Dab in Subarea 4 and Division 3.a. Official dab landings by country in Subarea 4 from 1950 – 2024.**

| Year | BEL | DEU | DNK  | FRA | FRO | GBR  | NLD  | NOR | SWE | Subarea 4 |
|------|-----|-----|------|-----|-----|------|------|-----|-----|-----------|
| 1950 | 254 | 92  | 900  | 139 | 0   | 2555 | 2031 | 0   | 0   | 5971      |
| 1951 | 462 | 114 | 1800 | 90  | 0   | 3503 | 2221 | 0   | 0   | 8190      |
| 1952 | 386 | 74  | 1562 | 227 | 0   | 2823 | 2904 | 0   | 0   | 7976      |
| 1953 | 357 | 58  | 1337 | 189 | 0   | 2591 | 1383 | 0   | 0   | 5915      |
| 1954 | 255 | 62  | 1666 | 177 | 0   | 2393 | 1099 | 0   | 0   | 5652      |
| 1955 | 305 | 92  | 2923 | 161 | 0   | 1993 | 1149 | 0   | 0   | 6623      |
| 1956 | 338 | 99  | 1766 | 138 | 0   | 1660 | 1368 | 0   | 99  | 5468      |
| 1957 | 336 | 73  | 1983 | 154 | 0   | 1785 | 1669 | 0   | 127 | 6127      |
| 1958 | 290 | 71  | 2320 | 175 | 0   | 1885 | 1517 | 0   | 84  | 6342      |
| 1959 | 285 | 93  | 1433 | 146 | 0   | 2011 | 1265 | 0   | 6   | 5239      |
| 1960 | 246 | 70  | 1833 | 154 | 0   | 1813 | 1052 | 0   | 0   | 5168      |
| 1961 | 227 | 67  | 1497 | 161 | 0   | 1734 | 916  | 0   | 0   | 4602      |
| 1962 | 205 | 54  | 1357 | 147 | 0   | 1524 | 795  | 0   | 0   | 4082      |
| 1963 | 306 | 40  | 1660 | 128 | 0   | 1481 | 1000 | 0   | 0   | 4615      |
| 1964 | 424 | 48  | 1612 | 672 | 0   | 1177 | 1049 | 0   | 0   | 4982      |
| 1965 | 432 | 64  | 1841 | 734 | 0   | 1099 | 1349 | 0   | 0   | 5519      |
| 1966 | 507 | 65  | 1589 | 719 | 0   | 1215 | 1767 | 0   | 0   | 5862      |
| 1967 | 384 | 77  | 659  | 716 | 0   | 1147 | 1341 | 0   | 0   | 4324      |
| 1968 | 334 | 57  | 861  | 350 | 0   | 877  | 1516 | 0   | 0   | 3995      |
| 1969 | 302 | 69  | 984  | 448 | 0   | 689  | 1630 | 0   | 0   | 4122      |
| 1970 | 338 | 71  | 1476 | 588 | 0   | 752  | 1958 | 0   | 0   | 5183      |
| 1971 | 409 | 46  | 1546 | 618 | 0   | 986  | 2941 | 0   | 0   | 6546      |

| Year | BEL | DEU | DNK  | FRA | FRO | GBR  | NLD  | NOR  | SWE | Subarea 4 |
|------|-----|-----|------|-----|-----|------|------|------|-----|-----------|
| 1972 | 638 | 46  | 1816 | 727 | 0   | 1057 | 3617 | 0    | 0   | 7901      |
| 1973 | 678 | 41  | 1899 | 873 | 0   | 1349 | 3638 | 1179 | 0   | 9657      |
| 1974 | 281 | 59  | 1168 | 310 | 0   | 1227 | 4101 | 0    | 0   | 7146      |
| 1975 | 600 | 45  | 944  | 418 | 0   | 992  | 4031 | 0    | 3   | 7033      |
| 1976 | 489 | 52  | 852  | 306 | 0   | 816  | 3402 | 0    | 0   | 5917      |
| 1977 | 652 | 70  | 743  | 371 | 0   | 907  | 3959 | 0    | 0   | 6702      |
| 1978 | 520 | 64  | 799  | 513 | 0   | 1038 | 3473 | 0    | 0   | 6407      |
| 1979 | 484 | 87  | 1366 | 630 | 0   | 951  | 4724 | 0    | 1   | 8243      |
| 1980 | 518 | 24  | 1376 | 639 | 0   | 777  | 5023 | 0    | 0   | 8357      |
| 1981 | 542 | 31  | 1968 | 447 | 0   | 737  | 4729 | 0    | 0   | 8454      |
| 1982 | 460 | 42  | 2356 | 594 | 0   | 1002 | 5111 | 0    | 0   | 9565      |
| 1983 | 541 | 49  | 4428 | 495 | 0   | 1034 | 5318 | 0    | 0   | 11865     |
| 1984 | 603 | 35  | 3438 | 486 | 0   | 920  | 0    | 0    | 0   | 5482      |
| 1985 | 509 | 24  | 3535 | 404 | 0   | 1030 | 0    | 0    | 0   | 5502      |
| 1986 | 445 | 34  | 1400 | 289 | 0   | 1036 | 0    | 0    | 1   | 3205      |
| 1987 | 514 | 36  | 1574 | 434 | 0   | 1373 | 0    | 0    | 0   | 3931      |
| 1988 | 697 | 72  | 1324 | 349 | 0   | 1221 | 3404 | 0    | 0   | 7067      |
| 1989 | 443 | 117 | 1280 | 223 | 0   | 1232 | 2521 | 0    | 0   | 5816      |
| 1990 | 416 | 162 | 1103 | 214 | 0   | 802  | 0    | 0    | 4   | 2701      |
| 1991 | 491 | 290 | 1160 | 258 | 0   | 1249 | 0    | 0    | 0   | 3448      |
| 1992 | 464 | 218 | 699  | 217 | 0   | 1049 | 0    | 0    | 0   | 2647      |
| 1993 | 548 | 493 | 1016 | 235 | 0   | 1017 | 0    | 0    | 0   | 3309      |
| 1994 | 397 | 626 | 1307 | 133 | 0   | 1398 | 0    | 0    | 0   | 3861      |
| 1995 | 410 | 0   | 1306 | 155 | 1   | 1993 | 0    | 0    | 0   | 3865      |
| 1996 | 527 | 718 | 1484 | 177 | 0   | 1928 | 0    | 0    | 0   | 4834      |
| 1997 | 507 | 945 | 1399 | 124 | 0   | 2284 | 0    | 0    | 0   | 5259      |
| 1998 | 757 | 796 | 1024 | 126 | 0   | 2085 | 7971 | 0    | 0   | 12759     |
| 1999 | 802 | 758 | 1101 | 0   | 0   | 1964 | 8651 | 0    | 0   | 13276     |
| 2000 | 684 | 892 | 785  | 124 | 0   | 1534 | 6527 | 49   | 0   | 10595     |

| Year  | BEL | DEU   | DNK  | FRA | FRO | GBR  | NLD  | NOR | SWE | Subarea 4 |
|-------|-----|-------|------|-----|-----|------|------|-----|-----|-----------|
| 2001  | 575 | 878   | 839  | 206 | 0   | 1368 | 5886 | 47  | 0   | 9799      |
| 2002  | 516 | 582   | 1126 | 228 | 0   | 1224 | 4951 | 51  | 0   | 8678      |
| 2003  | 396 | 642   | 1580 | 154 | 0   | 1204 | 4955 | 77  | 0   | 9008      |
| 2004  | 382 | 767   | 1136 | 121 | 0   | 1158 | 4989 | 55  | 0   | 8608      |
| 2005  | 372 | 1105  | 1128 | 121 | 0   | 1193 | 5352 | 131 | 0   | 9402      |
| 2006  | 369 | 1 149 | 949  | 129 | 0   | 1415 | 5071 | 107 | 0   | 9189      |
| 2007  | 436 | 526   | 634  | 194 | 0   | 1212 | 6313 | 118 | 0   | 9433      |
| 2008  | 372 | 375   | 670  | 160 | 0   | 847  | 5544 | 61  | 0   | 8029      |
| 2009  | 349 | 262   | 489  | 197 | 0   | 648  | 4588 | 29  | 0   | 6562      |
| 2010  | 337 | 365   | 523  | 179 | 0   | 724  | 5097 | 16  | 0   | 7241      |
| 2011  | 243 | 312   | 622  | 166 | 0   | 646  | 4808 | 30  | 0   | 6827      |
| 2012  | 455 | 253   | 421  | 126 | 0   | 665  | 4136 | 40  | 0   | 6096      |
| 2013  | 406 | 332   | 404  | 84  | 0   | 647  | 3314 | 26  | 0   | 5213      |
| 2014  | 304 | 282   | 253  | 72  | 0   | 505  | 2907 | 23  | 0   | 4346      |
| 2015  | 247 | 244   | 747  | 75  | 0   | 339  | 2500 | 10  | 0   | 4162      |
| 2016  | 321 | 244   | 932  | 75  | 0   | 372  | 2611 | 35  | 0   | 4590      |
| 2017  | 210 | 137   | 340  | 0   | 0   | 381  | 2027 | 35  | 0   | 3130      |
| 2018  | 313 | 185   | 710  | 0   | 0   | 418  | 2292 | 22  | 0   | 3940      |
| 2019  | 310 | 166   | 860  | 32  | 0   | 402  | 2418 | 85  | 0   | 4273      |
| 2020  | 171 | 188   | 558  | 25  | 0   | 381  | 2202 | 84  | 6   | 3615      |
| 2021  | 109 | 109   | 432  | 13  | 0   | 304  | 1901 | 28  | 6   | 2924      |
| 2022* | 107 | 49    | 317  | 7   | 0   | 241  | 947  | 13  | 0   | 1681      |
| 2023* | 75  | 50    | 336  | 7   | 0   | 191  | 560  | 12  | 6   | 1237      |
| 2024* | 112 | 50    | 178  | 5   | 0   | 232  | 641  | 11  | 1   | 1231      |

\* Preliminary catch statistics

Table 5.8. Dab in Subarea 4 and Division 3.a. Official dab landings in ICES Division 3.a from 1950 – 2024.

| Year | BEL | DEU | DNK  | FRA | NLD | NOR | SWE | GBR | Division 3.a |
|------|-----|-----|------|-----|-----|-----|-----|-----|--------------|
| 1950 | 0   | 34  | 1253 | 0   | 0   | 0   | 0   |     | 1287         |
| 1951 | 0   | 17  | 1315 | 0   | 0   | 0   | 0   |     | 1332         |

| Year | BEL | DEU | DNK  | FRA | NLD | NOR | SWE | GBR | Division 3.a |
|------|-----|-----|------|-----|-----|-----|-----|-----|--------------|
| 1952 | 0   | 21  | 1273 | 0   | 0   | 0   | 0   |     | 1294         |
| 1953 | 0   | 9   | 1114 | 0   | 0   | 0   | 0   |     | 1123         |
| 1954 | 0   | 4   | 1233 | 0   | 0   | 0   | 0   |     | 1237         |
| 1955 | 0   | 3   | 1254 | 0   | 0   | 0   | 0   |     | 1257         |
| 1956 | 0   | 5   | 1462 | 0   | 0   | 0   | 614 |     | 2081         |
| 1957 | 0   | 5   | 2025 | 0   | 0   | 0   | 694 |     | 2724         |
| 1958 | 0   | 4   | 1578 | 0   | 0   | 0   | 628 |     | 2210         |
| 1959 | 0   | 2   | 1307 | 0   | 0   | 0   | 634 |     | 1943         |
| 1960 | 0   | 1   | 1313 | 0   | 0   | 0   | 0   |     | 1314         |
| 1961 | 0   | 0   | 1367 | 0   | 0   | 0   | 0   |     | 1367         |
| 1962 | 0   | 2   | 1681 | 0   | 0   | 0   | 0   |     | 1683         |
| 1963 | 0   | 0   | 1565 | 0   | 0   | 0   | 0   |     | 1565         |
| 1964 | 0   | 1   | 1574 | 0   | 0   | 0   | 0   |     | 1575         |
| 1965 | 0   | 1   | 2051 | 0   | 0   | 0   | 0   |     | 2052         |
| 1966 | 0   | 0   | 1755 | 0   | 0   | 0   | 0   |     | 1755         |
| 1967 | 0   | 0   | 1115 | 0   | 0   | 0   | 0   |     | 1115         |
| 1968 | 0   | 0   | 1535 | 13  | 0   | 0   | 0   |     | 1548         |
| 1969 | 0   | 0   | 1430 | 0   | 0   | 0   | 0   |     | 1430         |
| 1970 | 0   | 0   | 1079 | 0   | 0   | 0   | 0   |     | 1079         |
| 1971 | 0   | 0   | 1242 | 0   | 0   | 0   | 0   |     | 1242         |
| 1972 | 0   | 0   | 1669 | 0   | 0   | 0   | 0   |     | 1669         |
| 1973 | 0   | 0   | 1449 | 0   | 0   | 0   | 0   |     | 1449         |
| 1974 | 0   | 0   | 2003 | 0   | 0   | 0   | 0   |     | 2003         |
| 1975 | 0   | 0   | 1959 | 0   | 2   | 0   | 88  |     | 2049         |
| 1976 | 10  | 0   | 1493 | 0   | 80  | 0   | 0   |     | 1583         |
| 1977 | 11  | 0   | 2105 | 0   | 142 | 0   | 60  |     | 2318         |
| 1978 | 2   | 0   | 2515 | 0   | 39  | 0   | 74  |     | 2630         |
| 1979 | 3   | 0   | 2616 | 0   | 15  | 0   | 82  |     | 2716         |
| 1980 | 3   | 0   | 2218 | 0   | 3   | 0   | 109 |     | 2333         |

| Year | BEL | DEU | DNK  | FRA | NLD | NOR | SWE | GBR | Division 3.a |
|------|-----|-----|------|-----|-----|-----|-----|-----|--------------|
| 1981 | 0   | 0   | 2574 | 0   | 5   | 0   | 100 |     | 2679         |
| 1982 | 1   | 0   | 2823 | 0   | 22  | 0   | 56  |     | 2902         |
| 1983 | 1   | 0   | 2759 | 0   | 34  | 0   | 112 |     | 2906         |
| 1984 | 0   | 0   | 2695 | 0   | 0   | 0   | 74  |     | 2769         |
| 1985 | 1   | 0   | 1486 | 0   | 0   | 0   | 58  |     | 1545         |
| 1986 | 5   | 0   | 1551 | 0   | 0   | 0   | 52  |     | 1608         |
| 1987 | 19  | 0   | 2182 | 0   | 0   | 0   | 57  |     | 2258         |
| 1988 | 13  | 0   | 2150 | 0   | 15  | 0   | 76  |     | 2254         |
| 1989 | 4   | 0   | 2302 | 0   | 0   | 0   | 40  |     | 2346         |
| 1990 | 3   | 0   | 1535 | 0   | 0   | 0   | 36  |     | 1574         |
| 1991 | 5   | 1   | 1556 | 0   | 0   | 0   | 47  |     | 1609         |
| 1992 | 10  | 0   | 1412 | 0   | 0   | 0   | 32  |     | 1454         |
| 1993 | 7   | 0   | 1656 | 0   | 0   | 0   | 32  |     | 1695         |
| 1994 | 9   | 0   | 1917 | 0   | 0   | 0   | 35  |     | 1961         |
| 1995 | 3   | 0   | 1482 | 0   | 0   | 0   | 45  |     | 1530         |
| 1996 | 0   | 0   | 1387 | 0   | 0   | 0   | 18  |     | 1405         |
| 1997 | 0   | 0   | 990  | 0   | 0   | 0   | 22  |     | 1012         |
| 1998 | 0   | 0   | 942  | 0   | 0   | 0   | 19  |     | 961          |
| 1999 | 0   | 0   | 661  | 0   | 0   | 0   | 12  |     | 673          |
| 2000 | 0   | 0   | 647  | 0   | 0   | 1   | 6   |     | 654          |
| 2001 | 0   | 0   | 751  | 0   | 0   | 7   | 7   |     | 765          |
| 2002 | 0   | 0   | 968  | 0   | 0   | 3   | 6   |     | 977          |
| 2003 | 0   | 0   | 674  | 0   | 173 | 14  | 4   |     | 865          |
| 2004 | 0   | 0   | 637  | 0   | 138 | 1   | 3   |     | 779          |
| 2005 | 0   | 0   | 738  | 0   | 95  | 0   | 3   |     | 836          |
| 2006 | 0   | 20  | 566  | 0   | 117 | 18  | 4   |     | 725          |
| 2007 | 0   | 9   | 547  | 0   | 126 | 3   | 9   |     | 694          |
| 2008 | 0   | 12  | 475  | 0   | 26  | 2   | 7   |     | 522          |
| 2009 | 0   | 4   | 478  | 0   | 3   | 1   | 12  |     | 498          |

| Year  | BEL | DEU | DNK | FRA | NLD | NOR | SWE | GBR | Division 3.a |
|-------|-----|-----|-----|-----|-----|-----|-----|-----|--------------|
| 2010  | 0   | 4   | 426 | 0   | 151 | 0   | 8   |     | 589          |
| 2011  | 0   | 10  | 517 | 0   | 0   | 11  | 7   |     | 545          |
| 2012  | 0   | 5   | 632 | 0   | 0   | 10  | 6   |     | 653          |
| 2013  | 0   | 11  | 654 | 0   | 174 | 26  | 6   |     | 871          |
| 2014  | 0   | 12  | 501 | 0   | 75  | 2   | 21  |     | 611          |
| 2015  | 0   | 8   | 752 | 0   | 203 | 8   | 24  |     | 995          |
| 2016  | 0   | 9   | 657 | 0   | 189 | 14  | 26  |     | 895          |
| 2017  | 0   | 3   | 601 | 0   | 157 | 14  | 13  |     | 788          |
| 2018  | 0   | 10  | 586 | 0   | 230 | 2   | 2   |     | 830          |
| 2019  | 0   | 1   | 675 | 0   | 390 | 1   | 2   |     | 1069         |
| 2020  | 0   | 1   | 457 | 0   | 174 | 0   | 3   | 2   | 637          |
| 2021  | 0   | 3   | 448 | 0   | 244 | 0   | 1   |     | 696          |
| 2022  | 0   | 0   | 449 | 0   | 52  | 0   | 1   |     | 502          |
| 2023* | 0   | 3   | 468 | 0   | 22  | 0   | 1   |     | 494          |
| 2024* | 0   | 1   | 408 | 0   | 85  | 0   | 1   |     | 496          |

\* Preliminary catch statistics

Table 5.9. Dab in Subarea 4 and Division 3.a. InterCatch landings, discards and total catch (2002–2024).

| Year | Landings | Imported discards | Raised discards | Total discards | Total catch | % discards |
|------|----------|-------------------|-----------------|----------------|-------------|------------|
| 2002 | 8588     | 14448             | 12183           | 26631          | 35219       | 76%        |
| 2003 | 9433     | 22152             | 22778           | 44930          | 54363       | 83%        |
| 2004 | 8647     | 18559             | 15714           | 34273          | 42920       | 80%        |
| 2005 | 9537     | 21295             | 13996           | 35291          | 44828       | 79%        |
| 2006 | 10236    | 16106             | 21871           | 37977          | 48213       | 79%        |
| 2007 | 9881     | 8936              | 24392           | 33328          | 43209       | 77%        |
| 2008 | 8645     | 14781             | 12598           | 27379          | 36024       | 76%        |
| 2009 | 7040     | 20652             | 12769           | 33421          | 40461       | 83%        |
| 2010 | 8279     | 23688             | 18798           | 42486          | 50765       | 84%        |
| 2011 | 7422     | 28227             | 16234           | 44460          | 51882       | 86%        |
| 2012 | 7047     | 33220             | 19412           | 52632          | 59679       | 88%        |



|      |      |       |       |        |       |     |
|------|------|-------|-------|--------|-------|-----|
| 2013 | 6611 | 36855 | 16621 | 53476  | 60087 | 89% |
| 2014 | 5048 | 35383 | 18350 | 53733  | 58781 | 91% |
| 2015 | 5082 | 26468 | 20904 | 47372  | 52454 | 90% |
| 2016 | 5085 | 29023 | 15788 | 44811  | 49896 | 90% |
| 2017 | 3598 | 22241 | 9274  | 31515  | 35113 | 90% |
| 2018 | 4233 | 28630 | 11915 | 40559* | 44792 | 91% |
| 2019 | 4991 | 26330 | 9372  | 35734* | 40725 | 88% |
| 2020 | 3808 | 22291 | 16575 | 38866  | 42674 | 91% |
| 2021 | 3321 | 20379 | 14031 | 34428* | 37749 | 91% |
| 2022 | 2171 | 21836 | 11679 | 35690* | 35690 | 94% |
| 2023 | 1802 | 15728 | 10464 | 26203* | 28005 | 94% |
| 2024 | 1716 | 14651 | 6803  | 21459* | 23174 | 93% |

\* including BMS landings

**Table 5.10. Dab in Subarea 4 and Division 3.a. Assessment summary. Weights are in tonnes. Landings are official estimates from 2000–2001. Landings, discards and catches are ICES estimates from 2002–2024. Catch C, biomass index I, harvest rate C/I and fishing pressure proxy relative to MSY proxy  $f$  are given for the years 2014–2022 used in the application of the *chr* advice method (ICES 2025). Relative recruitment R, and total mortality Z from SURBAR update.**

| Year | Land-ings | Dis-cards | Catch C | Index I | Harvest Rate<br>(C / I) | $f = L_{\text{mean}}/L_{F=M}$ | R age 1 | R low | R high | Z     | Z low | Z high |
|------|-----------|-----------|---------|---------|-------------------------|-------------------------------|---------|-------|--------|-------|-------|--------|
| 2000 | 11249     |           |         |         |                         |                               |         |       |        |       |       |        |
| 2001 | 10564     |           |         |         |                         |                               |         |       |        |       |       |        |
| 2002 | 8588      | 26631     | 35219   |         |                         |                               |         |       |        |       |       |        |
| 2003 | 9433      | 44931     | 54364   | 71.1    |                         |                               | 1.195   | 0.981 | 1.432  | 0.464 | 0.324 | 0.585  |
| 2004 | 8647      | 34273     | 42920   | 71.75   |                         |                               | 0.847   | 0.702 | 1.016  | 0.511 | 0.397 | 0.615  |
| 2005 | 9537      | 35291     | 44828   | 83.67   |                         |                               | 1.048   | 0.871 | 1.240  | 0.453 | 0.340 | 0.559  |
| 2006 | 10236     | 37977     | 48213   | 77.41   |                         |                               | 1.074   | 0.896 | 1.290  | 0.426 | 0.319 | 0.532  |
| 2007 | 9881      | 33328     | 43209   | 94.96   |                         |                               | 1.122   | 0.940 | 1.358  | 0.518 | 0.400 | 0.633  |
| 2008 | 8645      | 27379     | 36024   | 82.69   |                         |                               | 1.376   | 1.146 | 1.654  | 0.541 | 0.435 | 0.647  |
| 2009 | 7040      | 33421     | 40461   | 82.8    |                         |                               | 1.564   | 1.292 | 1.893  | 0.501 | 0.395 | 0.612  |
| 2010 | 8279      | 42486     | 50765   | 102.63  |                         |                               | 1.802   | 1.518 | 2.168  | 0.307 | 0.194 | 0.406  |
| 2011 | 7422      | 44460     | 51882   | 114.03  |                         |                               | 2.224   | 1.854 | 2.679  | 0.384 | 0.280 | 0.494  |
| 2012 | 7047      | 52632     | 59679   | 147.08  |                         |                               | 2.780   | 2.295 | 3.312  | 0.471 | 0.353 | 0.574  |

| Year | Land-ings | Dis-cards | Catch C | Index I | Harvest Rate<br>(C / I) | $f = L_{mean}/L_{F=M}$ | R age 1 | R low | R high | Z     | Z low | Z high |
|------|-----------|-----------|---------|---------|-------------------------|------------------------|---------|-------|--------|-------|-------|--------|
| 2013 | 6611      | 53476     | 60087   | 164.88  |                         |                        | 3.598   | 2.968 | 4.322  | 0.423 | 0.314 | 0.526  |
| 2014 | 5048      | 53733     | 58781   | 186.14  | 315.79                  | 1.07                   | 3.730   | 3.101 | 4.536  | 0.403 | 0.291 | 0.512  |
| 2015 | 5082      | 47372     | 52454   | 204.55  | 256.43                  | 1.09                   | 4.376   | 3.643 | 5.325  | 0.620 | 0.506 | 0.734  |
| 2016 | 5085      | 44811     | 49896   | 256.96  | 194.18                  | 1.09                   | 3.794   | 3.147 | 4.615  | 0.582 | 0.474 | 0.693  |
| 2017 | 3598      | 31515     | 35113   | 226.79  | 154.83                  | 1.08                   | 2.476   | 2.047 | 2.939  | 0.556 | 0.442 | 0.666  |
| 2018 | 4233      | 40559     | 44792   | 202.29  | 221.42                  | 1.08                   | 2.463   | 2.052 | 2.957  | 0.498 | 0.391 | 0.607  |
| 2019 | 4991      | 35734     | 40725   | 205.69  | 198                     | 1.07                   | 1.879   | 1.546 | 2.275  | 0.508 | 0.400 | 0.626  |
| 2020 | 3808      | 38866     | 42674   | 187.29  | 227.85                  | 1.06                   | 1.831   | 1.492 | 2.248  | 0.567 | 0.457 | 0.678  |
| 2021 | 3321      | 34428     | 37749   | 164.67  | 229.24                  | 1.07                   | 2.396   | 1.975 | 2.957  | 0.604 | 0.487 | 0.717  |
| 2022 | 2171      | 33519     | 35690   | 141.69  | 251.89                  | 1.05                   | 2.009   | 1.609 | 2.511  | 0.453 | 0.341 | 0.573  |
| 2023 | 1802      | 26203     | 28005   | 162.24  | 172.62                  | 1.05                   | 2.537   | 1.967 | 3.288  | 0.536 | 0.385 | 0.685  |
| 2024 | 1716      | 21459     | 23175   | 143.85  | 161.1                   | 1.05                   | 1.429   | 0.994 | 2.012  | 0.531 | 0.450 | 0.607  |

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